

iStartek GPS tracker Communication Protocol

Ver.: V2.3

www.istartek.com

Applied Models

The document describes the format of position/alarm GPRS data, and it is applied for the following models:

- ❖ PT60-L
- ❖ VT100 Series
- ❖ [VT110 Series](#)
- ❖ VT120-L
- ❖ [VT130-L](#)
- ❖ [VT150 Series](#)
- ❖ VT200 Series
- ❖ [VT300-L](#)
- ❖ VT800-L

Protocol modification record:

Version	Modify Content
V1.0	Initial version
V1.1	01 Added command and descriptions for 144, 212, 251, and 901. 02 Added 808 command to query parameters, support 100, 101, 102, 105, 106, 109 command
V1.2	01 Added the definition description of the index of the 211 command. 02 Modified the 252 command, remove the magnetic reader option settings. 03 Added 260, 261, 262, 263, 264, 265, 270 command functions, and add 56 event code. 04 139 command added option for 250cm Ultrasonic fuel sensor. 05 Added OBD-data data and OBD accessories. 06 Added the description of the calculation method of the data packet checksum.
V1.3	01 Added 142 instructions to set the data source of the fuel tank. 02 Modified the fuel tank and AD1, AD2 in the 138, 139, 140, and 141 command to have a non-one-to-one correspondence. You can set the data source through 142 command.
V1.4	01 Added 147, 154, 271, 803 command functions 02 Modified the description of the authorization time of the 144 instruction 03 Added 57 events
V1.5	01 Modified the description and definition of commands 211, 260, 261, 262, 263, 264, 265 02 Added the explanation of picture name for the 54 events in Appendix A 03 Modified the description of point 03 in the 800 instruction
V1.6	01 Add 280 command 02 Add WIFI settings for 252 commands
V1.7	01 Added FOTA process example (Appendix B) 02 Add instructions for 990, 991, 992, 993, 994, 995 commands
V1.8	01 Add 282 command, deleted the 700, 701, 702 instructions of the old version of PT60
V1.9	01 Add 148 command
V2.0	01 Add description of Bit8 in system-sta 02 Add 155, 804 command
V2.1	01 Add 283 command----20240403 02 Add VT120-L models ---20240516
V2.2	01 Add 030, 700 command 02 Modify the compatibility setting of the 122 command for heartbeat packet type
V2.3	01 Added commands 158/300/301/302/303/716/717/718/750/751 02 Added the GPRS message data field <BLE Peripheral Data>, supplemented the <Temp Sensor> data field, and added descriptions for Bluetooth sensor temperature and humidity data; 03 Supplemented Alarm Code 5/6/46/47 and added Bluetooth sensor index reporting; 04 Updated command 137 to add descriptions for temperature and humidity; 05 Added ignition mode command 716, used to configure the ignition mode to hardware ACC/voltage ignition/dynamic/static ignition; 06 Update command 128, 129, 130 and 131 for VT300-L

Content

1. Copyright and Disclaimer	7
2. GPRS Command Format	7
3. GPRS Event Data Format	7
4. SMS Command Format	14
5. SMS Event Data Format	15
6. Command List	15
000 – Ordinary Upload of Device Positioning Data Package	15
010 – Ordinary Upload of Device Positioning Data Package	16
020 – Compressed Upload of Device Positioning Data Package	16
030 – Simple Heartbeat Packet of Device	17
100 – Set the GPRS Parameters of Server 1	17
101 – Set up a Backup Server for Server 1	17
102 – Set the GPRS Time Interval of Server 1	18
103 – Set the Reliable Data Transmission Mode of Server 1	19
104 – Set Server 1 Data Sorting Transmission Mode	20
105 – Set the GPRS Parameters of Server 2	20
106 – Set the GPRS Time Interval of Server 2	21
107 – Set the Reliable Data Transmission Mode of Server 2	21
108 – Set Server 2 Data Sorting Transmission Mode	22
109 – Set APN	22
110 – Set Device ID	23
111 – Set DNS IP	23
120 – Set Track by Distance Interval	24
121 – Set Heading Change Report	24
122 – Set Heartbeat	24
123 – Set Speeding Alarm	24
124 – Set Sleep Mode	25
125 – Set a Circular Geo-fence	25
126 – Set Geo-fence Name	26
127 – Delete Circular Geo-fence	26
128 – Set Acceleration Alarm	27
129 – Set Deceleration Alarm	27
130 – Set Harsh Turning Alarm	28
131 – Set Impact Alarm	28
132 – Set Vibration (tow) alarm	29
133 – Set Idling Alarm (stop without ACC OFF)	30

134 – Set Fatigue Driving Alarm	30
135 – Set Temperature Sensor	30
136 – Set High and Low Temperature Alarm	31
137 – Get Temperature Value	32
138 – Set Fuel Tank Parameters	32
139 – Set Fuel Sensor Type	33
140 – Set Fuel Steal Alarm	34
141 – Set Low Fuel Alarm	34
142 – Set the Data Source of the Fuel Tank	35
144 – Authorize RFID/iButton	35
145 – Delete RFID/iButton	36
146 – Check Authorization Status of RFID/iButton Card	36
147 – Set RFID/iButton Authorized Ignition Time	37
148 – Set Authorized Ignition Time after Ignition Off.....	38
151 – Set Vibration Detection Sensitivity	38
152 – Set Arming/Disarming	38
153 – Set GSM Jamming Alarm	39
154 – Set Low Speed Alarm	40
155 – Set ACC Off not to Update GNSS position.....	40
158 – Set the AGPS function switch.....	40
200 – Set SMS Time Interval	41
201 – Set Authorized Phone Number	41
202 – Set Device Call Event	42
203 – Set the Authorized Number SMS Alarm Event	42
204 – Set the Characters for Event SMS Alarm	43
205 – Set Two-way Audio Events	43
206 – Set up Listening Events	44
210 – Set GPRS Alarm Event	44
211 – Set Photo Taking Event	45
212 – Set Output to Generate Output Events	45
250 – Set Input Working Mode	46
251 – Set Output Working Mode	46
252 – Set RS232 Port to Connect External Devices	47
260 – Set the Resolution when Taking Photo	47
261 – Set the Time Interval for the Camera to Take Photos	47
262 – Command to Take Photos	48
263 – Read the List of Photos	50
264 – Delete Photo	50

265 – Upload Photo	51
270 – Set vehicle Type	51
271 – OBD Command Transparent Transmission	52
280 – Set WiFi AP	52
282 – Set Working Mode (long standby device)	53
283 – Set Positioning Time after Waking Up from Power Saving Mode.....	54
300 – Add Bluetooth Peripheral.....	54
301 – Remove Bluetooth Peripheral.....	55
302 – Query Configured Bluetooth Peripherals.....	56
303 – Set Device Bluetooth Connection Password.....	56
600 – Set SMS Command Password	57
601 – Set SMS Time Zone	57
602 – Set SIM Card PIN Code	58
610 – Set Initial Mileage	58
611 – Clear GPRS/SMS Buffer	58
612 – Initialization Parameter	59
620 – Set Device LED Indicator Mode	59
700 – Log Upload.....	60
716 – Set Ignition Mode.....	60
717 – Driving Behavior Direction Calibration.....	61
718 – Driving Behavior Detection Mode Configuration.....	62
750 – Log Upload.....	62
751 – Static Drift Filter Settings.....	63
800 – Get Current Location	63
801 – Get Device SN, IMEI, Software Version	64
802 – Get Device Working Status	64
803 – Check SIM Card CCID	65
804 – Check SIM Card Balance.....	65
808 – Check Device Parameters	66
900 – Output Control.....	66
901 – Output Control (Mode Output)	67
910 – Restart Device or Module	67
990 – FOTA Upgrade Detection	68
991 – FOTA Upgrade Authorization	68
992 – FOTA Data Packet delivery	69
993 – FOTA Data Check	70



File Name	iStartek GPS Tracker Protocol	Ver.	2.3
Update	2025-10-18	Page	6

994 – Device Update FOTA Program	70
995 – Cancel FOTA	71
Appendix A – Alarm Event Codes and Alarm Characters	71
Appendix B – Example FOTA process	74

1. Copyright and Disclaimer

Shenzhen iStartek Technology Co., Ltd. All rights reserved Copyright © 2019 iStartek.

iStartek and  are registered trademarks of Shenzhen iStartek Technology Co., Ltd.

This manual cannot be copied for any purpose, disseminated in any way without the written authorization of iStartek.

iStartek shall not be liable for direct, indirect or all losses (including but not limited to economic losses, personal injuries, property and assets) caused by the use or inappropriate use of the product or documents.

2. GPRS Command Format

GPRS command packet format (from server to tracker):

\$\$<pack-no><pack-len>,<ID>,<cmd-code>,<cmd-data><checksum>\r\n

GPRS command packet format (from tracker to server):

&&<pack-no><pack-len>,<ID>,<cmd-code>,<cmd-data><checksum>\r\n

Description:

The comma (,) is a symbol used to separate each data item, and there is no space before and after the comma.

&&: It is the header of packet from tracker to server

\$\$: It is the header of packet from server to tracker

pack-no: Data packet number, 1 byte, calculated cyclically from 0x3A to 0x7E. When the platform receives the request instruction from the device, or the device receives the instruction sent by the platform, and needs to reply to the other party, the reply pack-no must be consistent with the received instruction.

pack-len: Data packet length, in decimal format, the data packet length includes the data of "<ID>,<cmd-code>,<cmd-data>" (note: include the comma before the ID).

ID: Device ID number, the default is the IMEI number of the device.

cmd-code: Command code.

cmd-data: The content of the command data, see the description of each command in the following protocol.

checksum: the checksum of the data packet, which is the lowest of the cumulative sum calculation result of all the data before the checksum (&&<pack-no><pack-len>,<ID>,<cmd-code>,<cmd-data>) One byte, expressed in a 2-digit hexadecimal string format.

For example, the cumulative result of the data packet "&&A20,021104023195429,800" is '&'+&'+A'+2'+0'+','+'0'+2'+1'+1'+0'+4'+0'+2'+3'+1'+9'+5'+4'+2'+9'+','+'8'+0'+0'=0x04DA, take the lowest byte as 0xDA, and use a 2-digit hexadecimal string as "DA".

\r\n: It is the ending character, <CR><LF>.

If not specified, multi-byte binary data in cmd-data use big endian format. i.e. Most significant byte first.

3. GPRS Event Data Format

&&<pack-no><pack-len>,<ID>,<cmd>,<alm-code>,<alm-data>,<date-time>,<fix_flag>,<latitude>,<longitude>,<sat-quantity>,<HDOP>,<speed>,<course>,<altitude>,<odometer>,<MCC|MNC|LAC|CI>,<CSQ-quanlity>,<system-sta>,<in-sta>,<out-sta>,<ext-V|bat-V|ad1-V|...|adn-V>,<procode>,<fule_liter>,<tempsensor><checksum>\r\n GPRS event data description:

Example:

&&A147,021104023195429,000,0,,180106093046,A,22.646430,114.065730,8,0.9,54,86,76,326781,460|0|27B3|0EA7,27,0000000F,02,01,04E2|018C|01C8|0000,1,0104B0,01013D|02813546\r\n

Bluetooth data example:

&&O285,869671077911490,000,0,,250815032420,A,22.678581,114.046028,26,0.5,0,0,72,0,460|1|7759|061E640B,25,003F,02,00,04A0|01A2|0000|0000,1,710000|92003C,0B010930|1401052F,,010162|02020164|03020064|05055511|06060078085300000E0464|0707007C04E10064|08085930001|090901003C64|0A0A53|0B0B095A1F025AFE\r\n

Item	Description	Example
pack-head	Data header, 2 bytes	&&

pack-no	Data packet number, 1 byte, calculated cyclically from 0x3A to 0x7E. When the platform receives the request instruction from the device, or the device receives the instruction sent by the	A
----------------	---	---

	platform, and needs to reply to the other party, the reply pack-no must be consistent with the received instruction	
pack-len	Data packet length, decimal character format, the length is sum of {,<ID>,<000,<alm-code>,<alm-data>,<date-time>,<fix_flag>,<latitude>,<longitude>,<sat-quantity>,<HDOP>,<speed>,<course>,<altitude>,<odometer>,<MCC MNC LAC CI>,<CSQ-quantity>,<status>,<in-sta>,<out-sta>,<ext-V bat-V ad1-V ... adn-V>,<pro-code>,<fule_level>,<temp-sensor>}	147
ID	Device ID number, composed of 3~16 numbers or English letters, the default is the IMEI number of device	021104023195429
cmd	GPRS command data type flag 000 is a normal data from tracker to server, and the platform does not need to reply to receive confirmation; 010 is a normal data from tracker to server, which requires the platform to reply to the receipt confirmation; 020 is a compressed data package from tracker to server, which requires the platform to reply and confirm receipt.	000
alm-code	Event code of the alarm, please refer to Appendix A	0
alm-data	Additional description of alarm events, please refer to Appendix A	Empty means the event has no additional description
date-time	GMT0 date & time, format: YYMMDDHHmmss 01 YY: Year, which is the value of the year minus 2000, 2 characters 02 MM: Month, 01-12, 2 characters 03 DD: day, 01--31, 2 characters 04 HH: hour, 00--23, 2 characters 05 mm: minutes, 00-59, 2 characters 06 ss: second, 00--59, 2 characters	180106093046 means 2018-1-6 9:30:46
fix_flag	GPS Location status indicator, A = valid, V = invalid	A
latitude	Latitude, the unit is degree, decimal format If it is south latitude, add-in front	22.646430
longitude	Longitude, the unit is degree, decimal format If it is west longitude, add-in front	114.065730
sat-quantity	Number of satellite signals received, decimal format	8=8 satellites
HDOP	Horizontal positioning accuracy, decimal format, the smaller the value, the more accurate the accuracy	0.9
speed	GPS speed, the unit is km/h, decimal format	54
course	The direction and angle of the movement, in degrees, in decimal format	86
altitude	Altitude, in meters, decimal format	76
odometer	The accumulated total mileage, in meters, in decimal format	326781
MCC MNC LAC CI	Base station information, ' ' is used to separate each data; MCC, MNC: Decimal format LAC, CI: Hexadecimal format	460 0 27B3 0EA7 MCC = 460; MNC = 0;

		LAC = 0x27B3; CI = 0x0EA7;
CSQ-quantity	GSM signal value, 0-31, decimal format	27
system-sta	The working status of tracker, in 8-bit hexadecimal format, each bit represents a working status: Bit0: GPRS connection status of IP1, 1=connected, 0=disconnected Bit1: GPRS connection status of IP2, 1=connected, 0=disconnected Bit2: GPS positioning status, 1=valid, 0=invalid Bit3: External power connection status, 1=connected, 0=disconnected Bit4: GPS antenna connection status, 1=connected, 0=disconnected Bit5: Stop status, 1=stop, 0=move Bit6: Armed state, 1=armed, 0=disarmed Bit7: RFID/iButton login status, 1=log in, 0=log out Bit8: Device shedding status, 1 = shedding, 0 = not shedding Bit9: Virtual Ignition (1 = Ignition On, 0 = Ignition Off) Bit10: Driving Behavior Calibration Completed (1 = Completed, 0 = Not Completed) Bit11-bit31: Reserved	0000000F
in-sta	Input status, 2-digit hexadecimal format, each bit represents a working status: bit[0]-input1 status, 1=triggered, 0=non-triggered bit[1]-input2 status, 1=triggered, 0=non-triggered bit[2]-input3 status, 1=triggered, 0=non-triggered etc.;	02 input2 is in the trigger state, other inputs are in the non-trigger state
out-sta	Output status, 2-digit hexadecimal format, each bit represents a working status: bit[0] – output1 status, 1=active, 0=inactive bit[1] – output2 status, 1=active, 0=inactive bit[2] – output3 state, 1=active, 0=inactive bit[7] – BLE Relay1 status, 1=active, 0=inactive etc.;	01 output1=active 81 Output 1 is active, and BLE Relay 1 is also active.
ext-V bat-V ad1-V ... adn-V	Voltage value, the value of V*100, in hexadecimal format; ‘ ’ is used to separate each data; ext-V: external power supply voltage value; bat-V: internal battery voltage value; ad1-V... and -V: is the voltage value of the corresponding AD input ports ad1... adn. Different devices support different numbers of AD input ports;	04E2 018C 01C8 0000 The external power supply voltage value is 0x04E2=1250, V=1250/100=12.50V; The internal battery voltage value is 0x018C=396, V=396/100=3.96V; The AD1 input voltage value is 0x01C8=456, V=456/100=4.56V; AD2 input voltage value is 0x0000=0, V=0/100=0V; The device supports two AD inputs.

pro-code	Extended protocol version number to distinguish different extended protocol functions. Decimal format.	1
-----------------	--	---

fule_liter	① Indicates the number of liters in a fuel tank. ② If the device is connected to multiple fuel level sensors, upload the values for multiple tanks. ③ This value is in hexadecimal format, with each tank's value separated by a ' ' character. ④ The number of liters in a fuel tank is represented by 6 hexadecimal characters. ⑤ The first character, if not 0, indicates that the sensor used in the current tank is a Bluetooth sensor and represents the Bluetooth index number of the binding. ⑥ The second character indicates the tank number (all characters are in HEX format). ⑦ The third to sixth characters are the number of liters * 10.		0104B0 01# Fuel tank: 0x04B0, the decimal value is 1200, the fuel volume is 120.0 liters The 138, 139, 140 and 142 commands must be used to set the fuel tank parameters to detect the liters. 1104B0 01# Fuel tank: 0x04B0, decimal value 1200. Binds to the fuel level sensor with Bluetooth index 1.
temp-sensor	① Represents the value of a temperature/temperature-humidity sensor. ② If the device is connected to multiple sensors, upload the values of multiple sensors. ③ The value is in hexadecimal format, and each sensor data set is separated by a ' ' symbol. Each sensor value is represented by a 6- or 8-bit hexadecimal character. ④ The first two characters are the sensor number. ⑤ The third through sixth characters are the temperature value. The most significant bit of the temperature value is the sign bit, and the value is the actual temperature value * 10. ⑥ If the sensor represented by the number supports humidity, the seventh and eighth characters are the humidity value. ⑦ Temperature and humidity sensors are currently only supported by Bluetooth sensors, and require a VT300 device with Bluetooth functionality.		01013D 028135 01#Temperature sensor: 0x013D, the highest bit is 0 for positive temperature value, decimal value is 317, 3176/10= 31.7°C 02#Temperature sensor: 0x8135, the highest digit is 1 for negative temperature value, the decimal value is 309 (0x0135), 309/10= 30.9, then -30.9°C 0B# temperature and humidity sensor, which is decimal 11, corresponds to the Bluetooth sensor index 1 (11-10). The highest bit of temperature is 0, which is a positive temperature value. The decimal value is 317, 317/10= 31.7°C, and the decimal value of humidity is 50.
OBD-data			
RPM ENGINE_LOAD MAF_FLOW INTAKE_PRESSURE INTAKE_TEMP THROTTLE COOLANT_TEMP INSTANT_FUEL FUEL_LEVEL	RPM	The speed per minute, the unit is r/min, a decimal string. It is empty when the data cannot be read.	800 Indicates that the engine speed is 800 revolutions per minute
	ENGINE_LOAD	Percentage of engine load. The unit is% (percent), a decimal string. It is empty when the data cannot be read.	40 Indicates that the engine load is 40%
	MAF_FLOW	Unit: g/s (gallons per second), decimal string. It is empty when the data cannot be read.	4 Indicates that the intake air flow is 4 gallons per second
	INTAKE_PRESSURE	The unit is kPa (kilopascal), a decimal string. It is empty	45 Indicates the absolute pressure

		when the data cannot be read.	of the intake pipe is 45 kPa
	INTAKE_TEMP	Unit: °C (Celsius), decimal string. It is the actual value +40°C. It is empty when the data cannot be read.	83 Indicates the intake air temperature 83-40=43°C
	THROTTLE	The unit is% (percent), a decimal string. It is empty when the data cannot be read.	79 Indicates that the throttle position is at 79%
	COOLANT_TEMP	Engine coolant temperature. The unit is °C (Celsius), a decimal string. It is the actual value +40°C. It is empty when the data cannot be read.	128 Indicates engine coolant 128-40=88°C
	INSTANT_FUEL	The unit is L/h. Decimal string. It is the value of L*10, which is accurate to one decimal place. It is empty when the data cannot be read.	35 Indicates instantaneous fuel consumption 3.5 liters/h
	FUEL_LEVEL	The amount of remaining fuel. Decimal string. The percentage after the value is the remaining fuel percentage. The number with L after the value is the remaining fuel volume in liters. It is empty when the data cannot be read.	80% means 80% fuel remaining 35L means the remaining 35L fuel volume
BLE Peripheral Data	Temperature and humidity sensor	Frame Format: <Index><Sensor Type><Battery Percent> Data Explanation: <Battery Percent>: 2 characters Indicates the remaining battery percentage	Example data: 010162 Index 1, type 1, temperature and humidity sensor; 0x62: indicates 98% battery.
	Door magnetic sensor	Frame Format: <Index><Sensor Type><Door Status><Battery Percent> Data Explanation: <Battery Percent>: 2 characters Battery remaining percentage <Door Status>: 2 characters Door status, 0 - Closed, 1 - Open	Example data: 02020164 Magnetic sensor type 2, index 2; 0x01: Indicates door open status; 0x64: Indicates 100% battery.
	Bluetooth relay	NA	
	Bluetooth heart rate sensor	Frame Format: <Index><Sensor Type><Heart Rate><Respiration Rate> Data Explanation:	Example data: 05055511 Index 5, type 5, Bluetooth heart rate sensor;

		<Heart Rate>: 2 characters, real-time heart rate, unit: pulse per minute <Respiration Rate>: 2 characters, respiratory rate, unit: pulse per minute	0x55: indicates a heart rate of 85 per minute; 0x11: indicates a respiratory rate of 17 per minute
	Bluetooth speed sensor	Frame Format: <Index> <Sensor Type> <Speed> <RPM> <ODO> <Battery Percent> Data Explanation: <Speed>: 4 characters, real-time speed Unit: 0.1 km/h If the speed is 0xSSSS and < 0x8000, it indicates forward movement. If the vehicle speed is 0xSSSS ≥ 0x8000, it indicates reverse movement, and the real speed is 0x10000-0xSSSS. <RPM>: 4 characters, real-time rotation speed, unit: 0.1 rpm <ODO>: 8 characters, mileage, unit: 1 meter <Battery Percent>: 2 characters, remaining battery percentage	Example data: 06060078085300000E0464 Index 6, type 6 Bluetooth speed sensor; 0x0078: indicates 12 km/h 0x0853: indicates speed of 213.1 rpm 0x00000E04: indicates mileage of 3588 m 0x64: indicates battery level of 100%
	Bluetooth capacitive fuel sensor (CPM212)	Frame Format: <Index> <Sensor Type> <Volume> <Fuel Height> <Human Detected> <Battery Percent> Data Explanation: <Volume>: 4 characters, unit:0.1L, <Fuel Height>: 4 characters, unit: 0.1mm <Human Detected>: 2 characters. 0x00: No human 0x01: Detected human <Battery Percent>: 2 characters Battery remaining percentage	Example data: 0707007C04E10064 Index 7, Type 7, Bluetooth capacitive fuel sensor; 0x007C: Indicates fuel volume 12.4L 0x04E1: Indicates fuel height 124.9mm 0x00: Indicates no human presence detected nearby 0x64: Indicates battery charge 100%
	Bluetooth alcohol sensor	Frame Format: <Index> <Sensor Type> <Alcohol Concentration> <Use Time>	Example data: 080805930001 Bluetooth alcohol sensor with index 8 and type 8;

		Data Explanation: <Alcohol Concentration>: 4 characters, unit: ppm, Real time alcohol concentration. Maximum 0x1388 equals to decimal 5000ppm. <Use Time>: 4 characters, unit: hour, It is calculated since first power on, maximum 2 years=17520 hours, then must change to new sensor.	0x0593: indicates the alcohol concentration is 1427 ppm 0x0001: indicates the sensor has been used for one hour
	Bluetooth capacitive fuel sensor (Escort TD BLE)	Frame Format: <Index> <Sensor Type> <Calibrated> <Fuel Level> <Battery Percent> Data Explanation: <Calibrated>: 2 characters It indicates whether the calibration table is currently enabled for the sensor. 0x00: Uncalibrated 0x01: Calibrated <Fuel Level>: 4 characters If the calibration table is enabled, the unit represented here will be liters. If the calibration table is not enabled, then this represents the original capacitance value. <Battery Percent>: 2 characters Battery remaining percentage	Example data: 090901003C64 Bluetooth capacitive fuel sensor with index 9 and type 9; 0x01: indicates calibrated 0x003C: indicates 60L 0x64: indicates 100% battery charge
	Bluetooth temperature and humidity sensor (Escort TH BLE)	Frame Format: <Index> <Sensor Type> <Battery Percent> Data Explanation: <Battery Percent>: 2 characters Battery remaining percentage	Example data: 0A0A53 Bluetooth temperature and humidity sensor with index 10 and type 10; 0x53: Indicates the battery level is 83%.
	Bluetooth tire pressure sensor	Frame Format: <Index> <Sensor Type> <Pressure> <Tire Temperature> <Pressure Status> <Battery Percent> Data Explanation: <Pressure>: 4 characters. Unit:	Example data: 0B0B095A1F025A Bluetooth tire pressure sensor with index 11 and type 11; 0x095A: indicates tire pressure of 239.4 kPa 0x1F: indicates tire temperature

	0.1kPa It indicates the current tire pressure. <Tire Temperature> : 2 characters.Unit: 1°C. It indicates the current tire temperature. <Pressure Status> : 2 characters. It indicates the current tire status: 00 - Normal 01 - Leakage 02 - Inflating 03 - Starting(Tire rotation) 04 - Power on(Sensor powered on for the first time) <Battery Percent> : 2 characters Battery remaining percentage	of 31°C 0x02: indicates inflated state 0x5A: indicates battery charge of 90%
	① Represents real-time data from configured Bluetooth peripherals. ② If the device has configured and scanned multiple sensors, it will upload the values of all sensors. The values are formatted as hexadecimal characters, with every two characters forming a hexadecimal value. The data between each sensor group is separated by a ' ' character. ③ The first two characters constitute the Bluetooth peripheral index <Index>. ④ The third and fourth characters constitute the Bluetooth peripheral type <Sensor Type>. The length of the subsequent data varies depending on the type (refer to the description of the type in command 300). ⑤ The Bluetooth fuel level sensor data will be updated in the <Fuel liter> data field. ⑥ The temperature and humidity data from the Bluetooth temperature and humidity sensor will be updated in the <Temp Sensor> data field.	
checksum	The checksum of the data packet, in 2-digit hexadecimal format, the accumulation and calculation result of all the data before the checksum {&&<pack-no><pack-len>,<ID>,000,<alm-code>,<alm-data>,<date-time>,<fix_flag>,<latitude>,<longitude>,<sat-quantity>,<speed>,<course>,<altitude>,<odometer>,<MCC MNC LAC CI>,<CSQ-quanlity>,<status>,<in-sta>,<out-sta>,<bat-ad ext-ad ad1 ... adn>,<pro-code>,<fule_data>,<temp-sensor>}. the lowest byte of the accumulation and calculation result of is expressed in a hexadecimal string.	46 The lowest byte that represents the result of the accumulation is 0x46, and it is represented by a hexadecimal string as "46"
\r\n	End of data packet, which is <CR><LF>	\r\n

4. SMS Command Format

SMS command format (data sent from mobile phone to tracker):

<password>,<cmd-code>,<cmd-data>

SMS command format (data is sent from tracker to mobile phone):

<cmd-code>,<cmd-data>

Description:

The comma (,) is a symbol used to separate each data, and there is no space before and after the comma.

password: SMS command password, 4 characters, can be composed of any numbers, letters, and symbols. The default is 0000.

cmd-code: command code.

cmd-data: command data content, see the description of each command in the following protocol.

If not specified, multi-byte binary data in cmd-data use big endian format. i.e. Most significant byte first

5. SMS Event Data Format

<Alarm_Head_String>,<date-time>,<fix_flag>,<speed>,<course>,<altitude>,<CSQ-quality>,http://maps.google.com/maps?q=<latitude>,<longitude>

GPRS Event data description:

Example:

SOS,082619 09:30:46,A,54,286,76,31,http://maps.google.com/maps?q=22.646430,114.065730

Item	Description	Example
Alarm_Head_String	The header name of the SMS alarm event, please refer to Appendix A. In and out of the fence event, add (fence name) after Alarm_Head_String	SOS Exit Fence(Home)
date-time	GMT0 + Time zone, date & time, format: MMDDYY hh:mm:ss 01 MM: month, 01-12, 2 characters 02 DD: day, 01--31, 2 characters 03 YY: year, which is the year minus 2000, 2 characters 04 hh: hour, 00--23, 2 characters 05 mm: minutes, 00-59, 2 characters 06 ss: second, 00--59, 2 characters 07 YY and hh are separated by a space symbol 08 hh, mm, ss are separated by ":"	082619 09:30:46=2019-08-26 09:30:46
fix_flag	GPS Location status indicator, A = valid, V = invalid	A
speed	GPS speed, the unit is km/h, decimal format	54
course	The direction and angle of the movement, in degrees, in decimal format	286
altitude	Altitude, in meters, decimal format	76
CSQ-quantity	GSM signal value, 0-31, decimal format	31
http://maps.google.com/maps?q=	Links located on Google Maps	http://maps.google.com/maps?q=
latitude	Latitude, in degrees, decimal format If it is south latitude, add - in front	22.646430
longitude	Longitude, in degrees, decimal format If it is west longitude, add - in front	114.065730

6. Command List

000 – Ordinary upload of device positioning data package

Supported Modes	GPRS (no need for platform to reply receipt confirmation)
Supported models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Server	See above GPRS event data packet format
Command Description	01 Upload the positioning data package, the device will automatically delete the data after uploading
Command Reply	No
Command Example	See above GPRS event data packet format
Remarks	

010 – Ordinary upload of device positioning data package

Supported Modes	GPRS (requires platform to reply receipt confirmation)
Supported models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Server	See above GPRS event data packet format
Command Description	01 The positioning data package is uploaded. After uploading, the platform must reply receipt confirmation before deleting the positioning data 02 The pack-no in the platform's reply must be the same as the pack-no in the uploaded positioning data package
Command Reply	010,1
Command Example	See above GPRS event data packet format
Remarks	

020 – Compressed upload of device positioning data package

Supported Modes	GPRS (requires platform to reply receipt confirmation)
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Server	See above GPRS event data packet format
Command Description	01 Multiple positioning data packages are compressed and uploaded. After uploading, the platform must reply receipt confirmation before deleting the positioning data 02 The pack-no in the platform's reply must be the same as the pack-no in the uploaded positioning data package
Command Reply	020,1
Command Example	See above GPRS event data packet format
Remarks	

030 – Simple Heartbeat Packet of Device

Supported Modes	GPRS
Supported Models	VT110 Series, VT130-L, VT150-L Series
Command to Server	&&<pack-no> <pack-len>,<ID>,<cmd> <checksum>\r\n
Command Description	01 Use the 122 command to configure the simple heartbeat packet upload interval. The default time interval is 0, and the simple heartbeat report is turned off. 02 The data packet is used to maintain the socket connection. If there is no data interaction within the set time interval, it will be reported once.
Command Reply	No
Command Example	&&?20,862771072953238,030EE
Remarks	

100 – Set the GPRS parameters of server 1

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Tracker	100,mode,IP,Port
Command Description	01 mode: GPRS connection mode, 0-Close GPRS connection, at this time any GPRS event will not be generated and sent to server;

	1-TCP connection mode; 2-UDP connection mode; 02 IP: The IP address or domain of the server, up to 32 characters; 03 Port: server port, decimal format, range value: 2~65534; 04 This command sets the IP and port of the main server, and 101 command sets the IP and port of the backup server. After the backup server is set, if the main server fails to connect (try to connect 5 times), the device will actively connect to the backup server (if it fails to connect to the backup server after 5 attempts, try to connect to the main server again).
Command Reply	100,<result> result: The result of Command Reply; OK-The command is set successfully; Unsupport- the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	100,1,device.istarmap.com,9094
Remarks	

101 – Set up a Backup Server for Server 1

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Tracker	101, IP,Port
Command Description	01 IP: The IP address or domain of the backup server, up to 32 characters; 02 Port: the port of the server, in decimal format, range value: 2~65534; 03 This command sets the IP and port of the backup server. After the backup server is set, if the main server fails to connect (try to connect 5 times), the device will actively connect to the backup server (if you try to connect to the backup server 5 times) If the connection fails, try to connect to the main server again); 04 The connection mode is subject to the mode set by 100 instructions; 05 The device will not connect to the main server and backup server at the same time.
Command Reply	101,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	101,87.172.73.51,9094
Remarks	

102 – Set the GPRS Time Interval of Server 1

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Tracker	102,normal_time,ACC_Off_time,stopping_time
Command	01 normal_time: Normal time interval. When the conditions of ACC_Off_time or stopping_time are

Description	not met, the device will upload GPRS positioning data at normal_time intervals; the unit is second, and the maximum is 65535 seconds. 02 ACC_Off_time: The time interval when ACC is off. When ACC_Off_time is set, normal_time is the time interval when ACC is on. The unit is seconds, and the maximum is 65535 seconds. 03 stopping_time: the time interval when the device stops. When stopping_time is set, the device will upload at normal_time intervals when the device is moving. The unit is seconds, and the maximum is 65535 seconds. 04 When both ACC_Off_time and stopping_time are set at the same time, the data will be uploaded according to the conditions of ACC_Off_time first.
Command Reply	102,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport- the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command Example	102,10 Set the normal time interval to 10 seconds; 102,10,60 Set the ACC on time interval to 10 seconds, and the ACC off time interval to 60 seconds; 102,10,,60 Set the normal time interval to 10 seconds when moving, and 60 seconds when stationary;
Remarks	

103 – Set the Reliable Data Transmission Mode of Server 1

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series, VT130-L, VT150 Series
Command to Tracker	103,mode
Command Description	01 mode: The method of reliable data upload, =0 means the device automatically deletes the data after the data is uploaded, =1 means the device needs the platform to reply to confirm data uploaded before deleting the data; the default is 0. 02 When selecting UDP to upload data, it is recommended to set the mode=1 transmission mode, because UDP is an unreliable transmission protocol. 03 When mode=0, the positioning data is uploaded with the 000 command; 04 When mode=1, the positioning data is uploaded with the 010 command, The receipt confirmation command returned by the platform also needs to be responded with the 010 command, and the pack-no must also be consistent with the pack-no of the positioning data packet.
Command Reply	103,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	103,0
Remarks	

104 – Set Server 1 Data Sorting Transmission Mode

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	104,mode
Command Description	01 mode: the method of sorting and uploading data, =0 means that real-time data is uploaded first when there is buffer, =1 means that all data is uploaded in FIFO mode; the default is 0. 02 When real-time tracking is required, it is recommended to set mode=0 transmission mode.
Command Reply	104,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command Example	104,0
Remarks	

105 – Set the GPRS Parameters of Server 2

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	105,mode,IP,Port
Command Description	01 mode: GPRS connection mode, 0-Close the GPRS connection, and no GPRS events will be sent to server 2 at this time; 1-TCP connection mode; 2-UDP connection mode; 02 IP: The IP address or domain of the server, up to 32 characters; 03 Port: the port of the server, in decimal format, range value: 2~65534; 04 When data needs to be uploaded to two servers at the same time, server 2 can be set. For example, when you need to upload data to your own server and government server.
Command Reply	105,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	105,1,device.istarmap.com,8012
Remarks	

106 – Set the GPRS Time Interval of Server 2

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	106,normal_time,ACC_Off_time,stopping_time
Command Description	01 normal_time: Normal time interval. When the conditions of ACC_Off_time or stopping_time are not met, the device will upload GPRS positioning data at normal_time intervals; the unit is second, and the maximum is 65535 seconds. 02 ACC_Off_time: The time interval when ACC is off. When ACC_Off_time is set, normal_time is the time interval when ACC is on. The unit is seconds, and the maximum is 65535 seconds. 03 stopping_time: the time interval when the device stops. When stopping_time is set, the device will upload at normal_time intervals when the device is moving. The unit is seconds, and the maximum is 65535 seconds. 04 When both ACC_Off_time and stopping_time are set at the same time, the data will be uploaded according to the conditions of ACC_Off_time first.

Command Reply	106,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	106,10 Set the normal time interval to 10 seconds; 106,10,60 Set the ACC on time interval to 10 seconds, and the ACC off time interval to 60 seconds; 106,10,,60 Set the normal time interval to 10 seconds when moving, and 60 seconds when stationary;
Remarks	

107 – Set the Reliable Data Transmission Mode of Server 2

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	107,mode
Command Description	01 mode: The method of reliable data upload, =0 means the device automatically deletes the data after the data is uploaded, =1 means the device needs the platform to reply to confirm data uploaded before deleting the data; the default is 0. 02 When selecting UDP to upload data, it is recommended to set the mode=1 transmission mode, because UDP is an unreliable transmission protocol. 03 When mode=0, the positioning data is uploaded with the 000 command; 04 When mode=1, the positioning data is uploaded with the 010 command, The receipt confirmation command returned by the platform also needs to be responded with the 010 command, and the pack-no must also be consistent with the pack-no of the positioning data packet.
Command Reply	107,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	107,0
Remarks	

108 – Set Server 2 Data Sorting Transmission Mode

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	108,mode
Command Description	01 mode: the method of sorting and uploading data, =0 means that real-time data is uploaded first when there is buffer, =1 means that all data is uploaded in FIFO mode; the default is 0. 02 When real-time tracking is required, it is recommended to set mode=0 transmission mode.

Command Reply	108,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	108,0
Remarks	

109 – Set APN

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	109,APN,APN_User_Name,APN_Password
Command Description	01 If APN does not have an APN username and APN password, leave it blank when setting; 02 Please confirm the correct SIM card APN information with the local network operator.
Command Reply	109,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	109,internet,internet,internet 109,CMNET
Remarks	

110 – Set Device ID

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	110,ID
Command Description	01 ID: 3-16 digits or English letters, the default is the 15-digit IMEI number of the device;
Command Reply	110,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	110,827643321798018 110,myID123456
Remarks	

111 –Set DNS IP

Supported Modes	GPRS/SMS
-----------------	----------

Supported Models	VT100 Series, VT200 Series
Command to Tracker	111,IP
Command Description	01 IP: The IP address of the DNS server, up to 16 characters;
Command Reply	111,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	111,121.236.189.173
Remarks	

120 – Set Track by Distance Interval

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	120,distance
Command Description	01 distance: The distance interval, in meters. The value range is 0~65535 meters, if it is 0, the function is turned off; the default is 0. 02 The recommended setting is not less than 100 meters. 03 When installed on a slow-moving vehicle, the track by distance interval function can be set to reduce the generation of GPRS traffic.
Command Reply	120,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	120,200 Set distance interval to 200 meters
Remarks	

121 – Set Heading Change Report

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	121,course
Command Description	01 course: The value of heading change, in degrees. The value range is 0~180 degrees, and it is 0 to turn off the function; the default is 0. 02 When the driving direction changes beyond the set value, a position information will be generated.

Command Reply	121,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	121,30 Set the heading change to 30 degrees
Remarks	

122 – Set Heartbeat

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	122,time,interval
Command Description	01 time: Heartbeat interval, in minutes. The value range is 0~65535 minutes. If it is 0, the function is turned off; the default is 60. 02 The heartbeat is only generated in the sleep state,and the heartbeat data is complete GPRS positioning data (event code is 31). 03 interval: simple heartbeat packet time interval, in minutes. The value range is 0~65535 minutes. If it is 0 or empty, the function is turned off; the default is empty; this time interval is only used in the simple heartbeat packet 030 data packet to maintain the connection with the platform.
Command Reply	122,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	122,60 Set the heartbeat interval to 60 minutes
Remarks	

123 – Set Speeding Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	123,speeding
Command Description	01 speeding: speeding alarm value, in km/h. The value range is 0~255 km/h, if it is 0, the function is turned off; the default is 0.
Command Reply	123,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	123,100 Set speeding alarm to 100 km/h

Remarks	
---------	--

124 – Set Sleep Mode

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	124,mode,wait_time
Command Description	01 mode: sleep mode, =0 to turn off sleep function, =1 to normal sleep mode, =2 to deep sleep mode; the default is 0. 02 wait_time: The device enters sleep mode after no input trigger, no vibration, no phone, no messages reaches the preset time, in seconds. If the sleep mode is set and the wait_time parameter is not set, the default is 300 seconds; 03 Normal sleep mode: GPS module power off, GSM module sleeps, it can be triggered by the input port, vibrate, call SMS activation; 04 Deep sleep mode: GPS module power off, GSM module power off, it can only be triggered by the input and vibration is activated.
Command Reply	124,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	124,1 Turn on the normal sleep mode, the time to enter the sleep mode is the default 300 seconds
Remarks	

125 – Set a Circular Geo-fence

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	125,index,flag,radius,latitude,longitude
Command Description	01 index: Geo-fence number, the value range is 1~8, up to 8 electronic fences; 02 flag: Alarm flag, =1 means exiting the fence alarm, =2 means entering the fence alarm, =3 means both exiting and entering the fence alarm; 03 radius: the radius of the circular Geo-fence, in meters, the maximum is 65535 meters; 04 latitude: the longitude of the center of the Geo-fence, in degrees; 05 longitude: the latitude of the center of the Geo-fence, in degrees; 06 When latitude, longitude are empty, use the current longitude and latitude of the device; 07 When the radius, latitude, and longitude are empty, use the current latitude and longitude of the device with a radius of 200 meters; 08 When flag, radius, latitude, longitude are empty, use the current location latitude and longitude of the device with a radius of 200 meters, which is an alarm for exiting the fence; 09 Set 06, 07, 08 when the GPS signal is valid, the setting can be successful, otherwise the setting will fail.

Command Reply	125,<result> result: the result of Command Reply;
	OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	125,1 125,1,1 125,1,1,200 The above 3 commands are all for setting the first Geo-fence with current location of the device in the longitude and latitude, the radius is 200 meters, exit the fence alarm; 125,1,1,200,22.540103,114.082329
Remarks	

126 – Set Geo-fence Name

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	126, index,name
Command Description	01 index: Geo-fence number, the value range is 1~8, up to 8 Geo-fences; 02 name: composed of 1~16 characters; 03 If name is set, it will be displayed as name in the SMS fence alarm event, see the above SMS event data format; 04 If the name is not set, it will be displayed as the fence number in the SMS fence alarm event.
Command Reply	126,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	126,1,Home
Remarks	

127 – Delete Circular Geo-fence

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	127,index
Command Description	01 index: Geo-fence number, the value range is 1~8, up to 8 Geo-fences; 02 When index is empty, delete all Geo-fences.
Command Reply	127,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command Example	127,1 Delete the first Geo-fence; 127 Delete all Geo-fences.
Remarks	

128 – Set Acceleration Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	128,acceleration
Command Description	01 acceleration: acceleration value, the unit is mg, the default is 450mg; when it is 0, the function is canceled. 02 When the acceleration exceeds acceleration, an emergency acceleration alarm is generated. VT300-L:(01-05 explanation only for VT300-L) 01. acceleration: Acceleration value, in mg. The default value is 0, indicating it is disabled. 02. When the acceleration exceeds acceleration, a sudden acceleration alarm is generated. 03. For bridge vehicles, we recommend setting 300 mg as a reference value and adjusting the sensitivity based on actual experience. 04. For large trucks, we recommend setting 270 mg as a reference value and adjusting the sensitivity based on actual experience. 05.The heavier the vehicle, the lower the trigger threshold is recommended for optimal sensitivity.
Command Reply	128,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	128,450
Remarks	

129 – Set Deceleration Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	129,deceleration
Command Description	01 deceleration: deceleration value, the unit is mg, the default is 800mg; when it is 0, the function is canceled. 02 When the deceleration exceeds the deceleration, an deceleration alarm is generated. VT300-L:(01-05 explanation only for VT300-L) 01. Deceleration: Negative acceleration value, in mg. The default value is 0, indicating it is disabled. 02. When the negative acceleration exceeds the deceleration value, a sudden deceleration alarm is generated. 03. For bridge vehicles, we recommend setting the value to 350 mg as a reference, and adjust the sensitivity based on actual experience.

	04. For large trucks, we recommend setting the value to 320 mg as a reference, and adjust the sensitivity based on actual experience. 05 For heavier vehicles, we recommend lowering the trigger threshold to optimize sensitivity.
Command Reply	129,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	129,700
Remarks	

130 – Set Harsh Turning Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	130,value
Command Description	01 value: harsh turning alarm value, the smaller the value, the more sensitive, the value range is [0,32], the default is 19; when it is 0, the function is canceled. 02 When the harsh turning exceeds preset value, a harsh turning alarm is generated. VT300-L:(01-05 explanation only for VT300-L) 01. value: Sharp turn detection threshold. Smaller values increase sensitivity. The value range is [0, 32]. The default value is 0, indicating it is disabled. 02. When the calculated sharp turn value exceeds this threshold, a sharp turn alarm is generated. 03. For bridge vehicles, it is recommended to set the value to 9 as a reference value, and adjust the sensitivity appropriately based on actual experience. 04. For large trucks, it is recommended to set the value to 6 as a reference value, and adjust the sensitivity appropriately based on actual experience. 05 The heavier the vehicle, the smaller the trigger threshold is recommended for optimal sensitivity.
Command Reply	130,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	130,17
Remarks	

131 – Set Impact Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	131,value
Command Description	01 value: impact sensitivity, the value range is 0~10, the smaller the value, the more sensitive, 0 is cancel function, and the default is 5;

	02 When using this function, it is recommended that the device be installed parallel to the horizontal plane. VT300-L:(01-05 explanation only for VT300-L) 01. Value: Collision sensitivity, ranging from 0 to 10, with lower values increasing sensitivity. 0 disables the function, and the default value is 0, indicating it is not enabled. 02. For bridge vehicles, we recommend setting a value of 8 as a reference, and adjust the sensitivity accordingly. 03. For large trucks, we recommend setting a value of 7 as a reference, and adjust the sensitivity accordingly. 04. Heavier vehicles require a lower trigger threshold for optimal sensitivity. 05 When using this function, it is recommended that the device be mounted parallel to a horizontal surface.
Command Reply	131,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	131,5
Remarks	

132 – Set Vibration (tow) alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	132,time,mode
Command Description	01 time: The duration of continuous vibration (movement). When the vibration time reaches the set value, an alarm will be generated. The value range is 0~255, the unit is second; =0 is to turn off the vibration alarm function, the default is 0; 02 mode: Check the conditions of vibration; =0, detect vibration under the condition of ACC OFF, the default is 0, when set to 0, the device needs to be connected to ACC detection. (PT60 does not have this item) =1, no need to detect vibration based on ACC. 03 If this function is used for vehicles, it is recommended to connect ACC and set mode=0. 04 The trailer tow will only be triggered when the tracker enters a static state (no vibration or movement within 3 minutes), and vibration occurs again.
Command Reply	132,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command	132,10
Example	132,10,1
Remarks	

133 – Set Idling Alarm (stop without ACC OFF)

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series
Command to Tracker	133,time
Command Description	01 time: Idling time, the value range is 0~255, the unit is minute; =0 close the function, the default is 0; 02 Idling condition: ACC is ON and the speed is 0, the tracker needs to be connected to ACC detection.
Command Reply	133,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	133,5 Alarm after 5 minutes of idling
Remarks	

134 – Set Fatigue Driving Alarm

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series
Command to Tracker	134,fatigue_time,rest_time,total_drive_time
Command Description	01 fatigue_time: single fatigue driving time, in minutes, 0 means to cancel the fatigue alarm function, the default is 0 minutes. When the time of parking after ACC OFF exceeds rest_time, the driving time is automatically cleared to 0. 02 rest_time: The minimum rest time, in minutes, the value range is [0, 255], and the default is 0 minutes; when the rest time exceeds rest_time, the driving time is automatically cleared to 0. 03 total_drive_time: total driving time. When the accumulated driving time in 24 hours exceeds total_drive_time, an fatigue driving alarm will be generated. 0 means to cancelthe fatigue driving function, and the default is 0 minutes;
Command Reply	134,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	134,240,20,480 Set the fatigue driving time to 240 minutes, the rest time to 20 minutes, and the cumulative driving time to 480 minutes
Remarks	

135 – Set Temperature Sensor

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	135,index,name
Command Description	01 index: the serial number of temperature sensor, 1~8, supports up to 8 sensors at the same time(1-WIRE type temperature sensor); 02 name: The name of the temperature sensor, up to 16 characters. 03 Installation of temperature sensor: A. Connect a temperature sensor, send the command 135, index, name, and the device will automatically configure the sensor as the sensor at the index position; B. Remove the previous sensor, connect another sensor, and send the command 135, index, name, and the device will automatically configure the sensor as the sensor at the index position; C. Repeat the above steps to set more sensors. Only one sensor can be connected at a time; 04 When the name is empty, delete the index sensor; 05 When both index and name are empty, delete all sensors.
Command Reply	135,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	135,1,Storage room Set 1# temperature sensor, named Storage room.
Remarks	

136 – Set High and Low Temperature Alarm

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	136,index,high_temp,low_temp
Command Description	01 index: the serial number of the temperature sensor, 1~8; 1-8: 1-Wire temperature sensors 11-22: Bluetooth temperature/humidity sensors (corresponding to Bluetooth indexes 1-12) 02 high_temp: high temperature alarm value, the unit is °C, accurate to one decimal place, if it is empty, cancel the high temperature alarm; 03 low_temp: low temperature alarm value, the unit is °C, accurate to one decimal place, if it is empty, the low temperature alarm will be canceled;
Command Reply	136,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	136,1,40.5,-10 Set 1# temperature sensor high temperature alarm to 40.5°C, low temperature alarm to -10°C.
Remarks	

137 – Get Temperature Value

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	137
Command Description	01Read all the configured sensor temperature values.
Command Reply	01 If the command is failed, reply: 137,<result> result: the result of Command Reply; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed. If the command is successful, reply: 137, <index1:Temp,index2:Temp,...,index8:Temp> <index11:Temp,Humidity,...index22:Temp,Humidity> index: The Nth sensor. If the response does not include the Nth sensor, it means that the Nth position is not configured. Temp: The temperature value in °C. If it is blank, it means that the temperature value of the sensor cannot be read. Humidity: The humidity percentage
Command Example	137 Reply: 137,1:29.0,2:28.8 Reply 2: 137, 1:29.0, 2:28.8, 11:29.0, 50, 15:28.9, 52
Remarks	Index is expressed in decimal;

138 – Set Fuel Tank Parameters

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT150
Command to Tracker	138,index,length,wide,high
Command Description	01 index: the serial number of the fuel tank, 1=1# fuel tank 2=2#fuel tank 02 length: The length of the fuel tank, in mm; the default is 0. 03 wide: the width of the fuel tank, it is the diameter of the circle if it is a cylindrical fuel tank,the unit is mm; the default is 0. 04 high: the height of the fuel tank, in mm; when high=0, it is a cylindrical fuel tank. The default is 0. 05 When length, wide, high, and name are all empty, delete the fuel sensor of the index.
Command Reply	138,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command	138,1,1000,500,600
Example	Set the length of 1# fuel tank to 1000mm, width to 500mm, and height to 600mm.
Remarks	

139 – Set Fuel Sensor Type

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT150
Command to Tracker	139,index,type,name
Command Description	01 index: the serial number of the fuel tank: 1=1# fuel tank 2=2#fuel tank 02 Fuel Sensor Type: 0=Not connect with fuel sensor, the default is 0. 1=Liquid level fuel sensor (Capactive fuel sensor, resistance fuel sensor, 0~5V output), 2=Ultrasonic fuel sensor (range 100CM, 0~5V output) 3=Ultrasonic fuel sensor (range 250CM, 0~5V output) 4=Escort capacitive fuel sensor (TD-600) 03 name: The name of the fuel tank, up to 16 characters.
Command Reply	139,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	139,1,1,Right Tank Set 1# fuel tank as the ultrasonic fuel sensor connected to AD1, and the name of the fuel tank is Right Tank.
Remarks	Type 3 ultrasonic fuel sensor supports Bluetooth as a data source. For details, refer to command 142

140 – Set Fuel Steal Alarm

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT150
Command to Tracker	140,index,theft-liter,time
Command Description	01 index: the serial number of the fuel tank: 1=1# fuel tank 2=2#fuel tank 02 theft-liter: The change in liters of the fuel stealing alarm. An alarm will be generated when the amount of fuel decreases within the time period and exceeds the set value. The unit is liter; 03 time: The time for check the change of the fuel quantity. An alarm will be generated when the fuel quantity decreases within the time period and exceeds the set value. The value range is [0,255], and the unit is second. 06 It is necessary to use commands 138, 139, 140 and 142 to set the parameters of the fuel tank to detect the alarm.

Command Reply	140,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	140,1,20,60 Set 1# fuel tank to generate fuel steal alarm when the reduction exceeds 20 liters within 60 seconds.
Remarks	

141 – Set Low Fuel Alarm

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT150
Command to Tracker	141,index,low-liter
Command Description	01 index: the serial number of the fuel tank, 1=1# fuel tank 2=2#fuel tank 02 low-liter: The number of liter changes for low fuel alarm. An alarm will be generated when the fuel volume is lower than the set value. Value range [0,255], the unit is liter; 01 It is necessary to use commands 138, 139, 141 and 142 to set fuel tank parameters to detect alarms.
Command Reply	141,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	141,1,10 Set 1# fuel tank to generate low fuel alarm when the fuel volume is less than 10 liters.
Remarks	

142 – Set the data source of the fuel tank

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT150
Command to Tracker	142,index,Source 142,index,Source, <Bind Index Of BLE> (This one is only for VT300-L)
Command Description	01 index: the serial number of the fuel tank, 1=1#fuel tank; 2=2#fuel tank; 02 Source: The data source of the fuel tank: 0=fuel value detected by AD1 1=fuel value detected by AD2 2=The fuel value read by RS232-1 3=The fuel value read by RS232-2 4 = Bluetooth fuel level sensor 03 Bind Index Of BLE: Bluetooth fuel level association index: Range: 1-12

	04 The default data source of 1# fuel tank is AD1, and the default data source of 2# fuel tank is AD2.
Command Reply	142,<result> result: Set the result of the command reply; OK-the command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Command Example	142,1,2 Set 1# fuel tank to use the fuel value read by RS232-1 to calculate the fuel volume and alarm for stealing fuel and low fuel.
Remarks	

144 – Authorize RFID/iButton

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	144,ID1,ID2,...,IDn
Command Description	01 ID[1,n] is the RFID/iButton card number, one command can authorize multiple ID numbers, ID number max 12 digits. 02 The RFID card number is a decimal string; the iButton card number is a hexadecimal string, the card number with a # sign in front is a hexadecimal number. 03 If only command 144 is sent and the card number is empty, it enters the manual card authorization mode. In this mode, the card number is automatically saved as authorization number after the card is swiped. Each time the card is swiped, it will automatically delay for 30 seconds to maintain the swiping authorization mode, and the mode will end after 30 seconds of not swiping card. Card swiping in this mode does not generate a card swiping event. We can send the 146 command to query the authorization status of the authorization card. For details, please refer to the 146 command description. 04 When the device uses the iButton swiping function, Output2 is fixed as the card swiping sound. A beep when swiping the iButton is detected. 05 While swiping card, OUT1 can be set to connect with relay to control the fuel and electricity. When you need OUT1 to control the fuel and electricity, you need to use the 212 command to set the 53 event to control OUT1. Swiping card to control fuel and electricity is a separate control logic, not according to the output mode set by the 251 command. 06 After set 53 event for output1 control, 53 events will not be uploaded when the unauthorized card is swiped, and 53 events will be uploaded when the authorized card is swiped. When the 53 event control output1 is canceled, any card swiping will upload 53 events. 07 After setting the 53 event control output of OUT 1, when ignition if the authorized card is not swiped (triggered by IN2), OUT1 will control to cut off immediately. 08 After swiping the authorization card, there is 60 seconds to ignite. If there is no ignition for more than 60 seconds, you need to swipe the authorization card again to ignite. 09 When the flame is turned off, there is 60 seconds to re-ignite. If the ignition is not performed for more than 60 seconds, you need to swipe the authorization card again to ignite.
Command Reply	144,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command Example	144,123456789,987654321 Authorization ID number 123456789,987654321 144,#00000A678C,#00000E4321F Authorization ID number 0x00000A678C, 0x00000E4321F 144 Enter manual swiping authorization mode, the device will reply "Under authorization" first, and then reply OK after 30 seconds without swiping the card
Remarks	

145 – Delete RFID/iButton

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	145,ALL/ID1,ID2,...,IDn
Command Description	01 ID[1,n] is the RFID/iButton card number, one command can delete multiple ID numbers. 02 The RFID card number is a decimal string; the iButton card number is a hexadecimal string, and the card number with a # sign in front is a hexadecimal number. 03 145, ALL, means delete all authorization cards 04 If only command 145 is sent and the card number is empty, it will enter the manual card deletion mode. In this mode, the card number will be deleted automatically after swiping the card. Each time swipe card, it will automatically delay 30 seconds to maintain the swipe delete mode, and the mode will end after 30 seconds of no swipe. Card swiping in the swipe delete mode does not generate a swiping event. 05 When the device uses the iButton swiping function, output2 is fixed as the card swiping sound (a buzzer should be connected to output 2) . A beep when swiping the iButton is detected.
Command Reply	145,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed. Deleting-Authorizing
Command	145,123456789,987654321
Example	Delete the ID number 123456789,987654321 145,#00000A678C,#00000E4321F Delete ID number 0x00000A678C, 0x00000E4321F 145,ALL Delete all authorization cards 145 Enter manual swipe delete mode, the device will reply "Delete" first, and then reply OK after 30 seconds without swiping the card
Remarks	

146 – Check Authorization Status of RFID/iButton Card

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L

Command to Tracker	146,ID1,ID2,...,IDn
Command Description	01 ID[1,n] is the card number of RFID/iButton, one command can query multiple ID numbers. 02 The RFID card number is a decimal string; the iButton card number is a hexadecimal string, and the card number with a # sign in front of it is a hexadecimal number.
Command Reply	146,<result> result: Set the result of the command reply; ID1:Flag1,ID2:Flag2,...,IDn: Flagn-Flag is status, 0 is not authorized, 1 is authorized; Unsupport-the instruction is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Command Example	146,123456789,987654321 Query ID number 123456789,987654321, reply 146,123456789:0,987654321:1 146,#00000A678C,#00000E4321F Query ID number 0x00000A678C,0x00000E4321F, reply 146,#00000A678C:1,#00000E4321F:1
Remarks	

147 – Set RFID/iButton Authorized Ignition Time

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	147,time
Command Description	01 time: The time period of ignition ON after swiping the authorized card. If the ignition on fails in this time period, you need to swipe the authorized card again. 02 The unit is second, the value range is [0, 255], and the default is 60 seconds.
Command Reply	147,<result> Result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	147,30 Set the authorized ignition time period to 30 seconds
Remarks	

148 – Set Authorized Ignition Time after Ignition Off

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	148,time
Command Description	01 time: The time after ignition off to keep the authorization to ignite. If not ignite engine in this time, you need to swipe the card again to ignite. 02 The unit is second, the value range is [0, 255], and the default is 5 seconds.
Command Reply	148,<result> Result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported;

	Failed-The command format or password is incorrect and the setting failed.
Command Example	148,30 Set the authorized ignition time after ignition off to 30 seconds
Remarks	

151 – Set Vibration Detection Sensitivity	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	151,sen
Command Description	01 sen: Sensitivity of vibration detection. The value range is [3,255], the smaller the value, the more sensitive it is. The default is 3.
Command Reply	151,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	151,3 Set sensitivity as 3
Remarks	

152 – Set Arming/Disarming	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,,VT110Series
Command to Tracker	152,flag
Command Description	01 flag: =0 means disarming, =1 means arming. The default is 0. 02 In the arming state, Input trigger (open door, ignition, etc.), towing, GSM interference will generate a car theft alarm. 03 The arming alarm function is independent of other events such as input port triggering and towing. The use of other events is not affected by arming or disarming.
Command Reply	152,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	152,1 Armed
Remarks	

153 – Set GSM Jamming Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,,VT110Series
Command to Tracker	153,time
Command Description	01 time: the value is [0,255] seconds, =0 is to cancel the GSM jamming detection alarmfunction, the default is 0. 02 When GSM is interfered for longer than the set time, an jamming alarm will be generated.
Command Reply	153,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	153,10 Set the jamming time to alarm after 10 seconds
Remarks	

154 – Set low speed alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,,VT110Series
Command to Tracker	154,speeding
Command Description	01 speeding: Low speed alarm threshold, the unit is km/h. The value range is 0~255 km/h, if it is 0, the function is turned off; The default is 0. 02 Trigger alarm only when ACC is ON and the speed is lower than the threshold.
Command Reply	154,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	154,10 Set the low speed alarm as 10 km/h
Remarks	

155 – Set ACC Off not to Update GNSS position

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT200 Series, VT300-L, VT800-L, VT120-L, VT110 Series
Command to Tracker	155,flag
Command Description	01 flag: =1 means the position will not be updated in ACC Off state, =0 means this function is disabled; the default value is 0.

Command Reply	155,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	155,1 Set the position not to be updated in ACC Off state
Remarks	

158 – Set the AGPS function switch

Supported Modes	GPRS/SMS
Command to Tracker	VT110 Series, VT130-L, VT150 Series, VT300-L
Supported Models	158,flag
Command Description	01 flag: =1 turns on the AGPS function, =0 turns it off; the default value is 1.
Command Reply	158,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	158,1 Enable AGPS.
Remarks	

200 – Set SMS Time Interval

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	200,normal_time, ACC_Off_time,stopping_time
Command Description	01 normal_time: Normal time interval. When the conditions of ACC_Off_time or stopping_time are not met, the device will upload GPRS positioning data at normal_time intervals; the unit is minute, and the maximum is 65535 minutes. 02 ACC_Off_time: The time interval when ACC is off. When ACC_Off_time is set, normal_time is the time interval when ACC is on. The unit is minute, and the maximum is 65535 minutes. 03 stopping_time: the time interval when the device stops. When stopping_time is set, the device will upload at normal_time intervals when the device is moving. The unit is minute, and the maximum is 65535 minutes. 04 When both ACC_Off_time and stopping_time are set at the same time, the data will be uploaded according to the conditions of ACC_Off_time first
Command Reply	200,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported;

	Failed-The command format or password is incorrect and the setting failed.
Command Example	Set the SMS time interval to 10 minutes; 200,10,60 Set the ACC on SMS time interval to 10 minutes, and the ACC off SMS time interval to 60 minutes; 200,10,,60 Set the SMS time interval to 10 minutes when moving, and the SMS time interval to 60 minutes when stationary;
Remarks	

201 – Set Authorized Phone Number

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	201,num1,num2,num3
Command Description	01 num1,num2,num3: 01 num1,num2,num3: authorized mobile phone numbers, up to 3 authorized numbers can be set; 02 The mobile phone number is up to 16 digits. 03 The authorized number can receive SMS alarm and phone alarm. 04 After setting the authorization number, the default event selected is event number 05 Delete all authorized numbers when only 201 command is issued
Command Reply	201,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	201,13412345678,13587654321,13856789123
Remarks	

202 – Set Device Call Event

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	202,index,operation,code1,code2,...
Command Description	01 index: the position number of the authorized number, 1 is the first number, 2 is the second number, and 3 is the third number; 02 operation: operation code, 1 is setting, 2 is adding, 3 is deleting; 03 code: 0=automatically answer, 1=automatically monitor, 2=automatically hang up the unauthorized code, 3=automatically reply location information, each event is separated by ' '; 04 Priority level: Auto hang up > Auto answer > Auto monitor

	05 is the corresponding processing method of the device when the mobile phone makes a call to the device.
Command Reply	202,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	202,1,1,0,3 Set up automatic answering and automatic reply to the first authorized number incoming call.
Remarks	

203 – Set the Authorized Number SMS Alarm Event

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	203,index,operation,code1,code2,...
Command Description	01 index: the position number of the authorized number, 1 is the first number, 2 is the second number, and 3 is the third number; 02 operation: operation code, 1 is setting, 2 is adding, 3 is deleting; 03 code: event code, see Annex A event code table, each event is separated by ' '; 04 When the set event is generated, the SMS alarm event will be sent to the corresponding authorized number.
Command Reply	203,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	203,1,2,2,7,15,20,39 Add the first authorized number to receive alarms for events 2, 7, 15, 20, 39
Remarks	

204 – Set the Characters for Event SMS Alarm

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	204,code,string
Command Description	01 code: event code, see Annex A event code table; 02 string: The description character of the SMS alarm, the longest is 32 characters; 03 The default alarm characters are shown in the event code table.
Command Reply	204,<result> result: the result of Command Reply; OK-The command is set successfully;

	Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	204,1,help Set the alarm character of the first event code to "help" .
Remarks	

205 – Set Two-way Audio Events

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	205,index,operation,code1,code2,...
Command Description	01 index: the position number of the authorized number, 1 is the first number, 2 is the second number, and 3 is the third number; 02 operation: operation code, 1 is setting, 2 is adding, 3 is deleting; 03 code: event code, see Annex A event code table, each event is separated by ' ' ; 04 When the set event occurs, a call will be made to the corresponding authorized number for two-way conversation.
Command Reply	205,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	205,1,1,2,7,15,20,39 Set the first authorized number to receive two-way calls for events 2, 7, 15, 20, 39
Remarks	

206 – Set up Listening Events

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	206,index,operation,code1,code2,...
Command Description	01 index: the position number of the authorized number, 1 is the first number, 2 is the second number, and 3 is the third number; 02 operation: operation code, 1 is setting, 2 is adding, 3 is deleting; 03 code: event code, see Annex A event code table, each event is separated by ' ' ; When the event set in 04 is generated, a call will be made to the corresponding authorized number to monitor the device.
Command Reply	206,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	206,1,1,2,7,15,20,39 Set the first authorized number to receive monitoring of events 2, 7, 15, 20, and 39

Remarks	
---------	--

210 – Set GPRS Alarm Event

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	210,index,operation,code1,code2,...
Command Description	01 index: server number, 1 is the GPRS event of server 1, and 2 is the GPRS event of server 2 02 operation: operation code, 1 is setting, 2 is adding, 3 is deleting 03 code: event code, see Annex A event code table, each event is separated by ' ' ; 04 GPRS alarm event will be generated when the set event is triggered.
Command Reply	210,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	210,1,1,2,7,15,20,39 Set the GPRS alarm of server 1 for events 2, 7, 15, 20, 39
Remarks	

211 – Set Photo Taking Event

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	211,index,operation,code1,code2,...
Command Description	01 index: camera number, 1 is camera 1, 2 is camera 2, 3 is camera 3; 02 The camera with the smallest serial number is 1# camera, which increases in order; for example, the 252 command sets the RS232-2 and RS232-4 to connect the camera, then the RS232-2 connects to the 1# camera, and the RS232-4 connects to the camera. It is the 2# camera. 03 operation: operation code, 1 is setting, 2 is adding, 3 is deleting 04 code: event code, see Annex A event code table, each event is separated by ' ' ; 05 A photo will be taken when the set event is generated.
Command Reply	211,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	211,1,1,2,7,15,20,39 Set up the No. 1 camera to take photos when events 2, 7, 15, 20, 39 triggered
Remarks	

212 – Set Output to Generate Output Events

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series, VT130-L, VT150 Series
Command to Tracker	212,index,operation,code1,code2,...
Command Description	01 index: output port number, 1 is output port 1, 2 is output port 2, and 3 is output port 3; 02 operation: operation code, 1 is setting, 2 is adding, 3 is deleting 03 code: event code, see Annex A event code table, each event is separated by ' ' ; 04 will generate output when the preset event is generated, and generate output according to the mode set by the 251 command. 05 Event 0, 17, 20, 23, 24, 27, 28, 29, 30, 31, 32, 33, 34, 36, 37, 54 do not support event output. 06 After events 1, 3, 5, 7, 9, 11, 18, 21, 22, 25, 26, 43, 45, 46, 47, 49, 50 trigger output, when the trigger source disappears, the output will return to its original state . 07 Please set this feature carefully.
Command Reply	212,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	212,1,1,2,7,15,20,39 Set output 1 for events 2, 7, 15, 20, 39 to generate output.
Remarks	

250 – Set Input Working Mode

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series
Command to Tracker	250,input,mode,time
Command Description	01 input: the serial number of the input port, 1=input1, 2=input2, 3=input3, 4=input4, 5=input5, 6=input6, and so on; 02 mode: input port working mode, 0=low level trigger, 1=high level trigger, 2=AD analog input; 03 time: The trigger time of the input port. The trigger event will be generated when the trigger time exceeds the set value. It is only valid for the high and low level trigger input modes, the unit is ms, the value is [0, 65535], when time =0 It is triggered immediately and the default is 0. Note: IN1 is fixed to low level trigger and cannot be set. IN2 is fixed as high level trigger and cannot be set. IN3 only supports high and low level trigger mode settings. IN4 only supports high and low level trigger mode settings. IN5 (AD1) only supports high-level trigger and AD input mode settings. IN6 (AD2) only supports high-level trigger and AD input mode settings. 04 The input modes for different models is different. 05 VT200 and VT200L series:

	IN1 supports high and low level trigger modes, and the default is low level trigger. IN2 supports high and low level trigger modes, and the default is high level trigger. IN3 supports three modes of high and low level trigger and AD2 (0~6.6V) input, and the default is high level trigger. IN4 (AD1) supports high-level trigger and AD input mode settings, and the default is AD input mode. 06 VT140: IN1 supports high and low level trigger modes, and the default is low level trigger. IN2 supports high and low level trigger modes, and the default is high level trigger. IN3 supports high and low level trigger modes, and the default is high level trigger. IN4 supports high and low level trigger modes, and the default is high level trigger. IN5 (AD1) supports two mode settings of high-level touch and AD input, and the default is AD input mode. IN6 (AD2) supports two mode settings of high level touch and AD input, and the default is AD input mode.
Command Reply	250,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	250,2,1,3000 Set input2 to high level trigger mode, the trigger time is 3000ms
Remarks	

251 – Set Output Working Mode

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L, VT110 Series
Command to Tracker	251,output,mode,time,on_time,off_time,safe_speed

Command Description	01 output: the serial number of the output port, 1= output1, 2= output2, 3= output3, and so on; 02 mode: output action mode, mode=1, turn on the output (the output is in a low-level state) mode, which is a long-time on state; mode=2, close the output (the output port is in a floating state) mode, which is a long-term off state; mode=3, pulse output (pulse output is generated with on_time and off_time parameters, when on_time and off_time are 0, turn the current output status into a flip change output); 03 time: The time when the output is generated, the unit is second, the value range is 0~65535; =0 is the infinite time output; 04 on_time: The time to turn on the output (the output port is in a low-level state), the unit is ms, and the value range is 0~65535; 05 off_time: The time to turn off the output (the output port is in a floating state), the unit is ms, and the value range is 0~65535; 06 When on_time and off_time are both 0, it is long output mode. 07 safe_speed: The speed limit of the action generated by the output port, the unit is km/h, and the value range is 0~255; When safe_speed=0 or empty, the output port immediately generates output action; When safe_speed>0, the output action is only generated when the current GPS speed is less than safe_speed and in the valid positioning state
Command Reply	251,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	251,1,3,10,500,500 The output 1 mode is 10 pulse output, and the pulse is a cycle of 500ms ON and 500ms OFF.
Remarks	

252 – Set RS232 Port to Connect External Devices

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	252,index,accessory
Command Description	01 index: Serial number, 1= RS232-1, 2= RS232-2, 3= RS232-3, and so on; 02 accessory: the external devices, 0=Empty 1=RFID card reader; 2=OBD decoder; 3=Camera; (VT300-L does not support this accessory) 4=Ultrasonic fuel feel; 5=WIFI modem; (VT300-L does not support this accessory) 03 The default is that RS232-1 is the RFID reader, and RS232-2 is empty.
Command Reply	252,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command Example	252,1,2 Set 1# RS232 connect with OBD Reader.
Remarks	

260 – Set the Resolution when taking photo

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	260,index,resolution
Command Description	01 index: camera serial number, 1= 1# camera, 2= 2# camera, 3= 3# camera, 4= 4# camera; maximum support 4 cameras. 02 The camera with the smallest serial number is 1# camera, which increases in order; for example, the 252 command sets the RS232-2 and RS232-4 to connect the camera, then the RS232-2 connects to the 1# camera, and the RS232-4 connects to the camera. It is the 2# camera. 02 resolution: The resolution of the photo, defined as follows, the default is 2. 1: 160*128 2: 320*240 3: 640*480
Command Reply	260,<result> Result: Set the result of the command reply; OK-the instruction is set successfully; Unsupport-the instruction is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Remarks	

261 – Set the time interval for the camera to take photos

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	261,index,normal_time,ACC_Off_time,stopping_time
Command Description	01 index: camera serial number, 1= 1# camera, 2= 2# camera, 3= 3# camera, 4= 4# camera; maximum support 4 cameras. 02 normal_time: Normal time interval for taking photos. When the conditions of ACC_Off_time or stopping_time are not met, the device will take photos at normal_time intervals; the unit is minutes, and the maximum is 65535 minutes. 03 ACC_Off_time: The time interval for taking photos when ACC is off. When ACC_Off_time is set, normal_time is the time interval when ACC is on. The unit is minute, and the maximum is 65535 minutes. 04 stopping_time: The time interval for taking photos when the device stops. When stopping_time is set, the upload will be performed at normal_time intervals when the device is moving. The unit is minute, and the maximum is 65535 minutes. 05 When both ACC_Off_time and stopping_time are set at the same time, take photos with ACC_Off_time first. 06 The event trigger source for timing photo is 0 event. 07 VT200 Series devices support storage of up to 128 photos, and when the number of photos exceeds 128, the oldest photos will be overwritten.

Command Reply	261,<result> result: Set the result of the command reply; OK-the instruction is set successfully; Unsupport-the instruction is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Examples	261,1,20 Set the time interval of 1# camera to take photos as 20 minutes; 261,1,20,60 Set the time interval of ACC on timed photographing to 20 minutes, and the time interval of ACC off timed photographing to 60 minutes; 261,1,20,,60 Set the time interval of timed photographing when moving is 20 minutes, and the time interval of timed photographing when stationary is 60 minutes;
Remarks	

262 – Command to take photos

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	262,ID1,ID2,ID3,ID4
Command Description	01 ID: camera serial number, 1= 1# camera, 2= 2# camera, 3= 3# camera, 4= 4# camera; maximum 4 cameras are supported. 02 One command can control one or more cameras to take photos. 03 If the IDn photo is successful, the reply IDn: OK, if the photo fails, the reply IDn: Err 04 When the device responds to the 262 command, a photo event is generated at the same time. The trigger source of the photo command is 34 events. The photo event data includes the photo name. 05 VT200 Series devices support storage of up to 128 photos, and when the number of photos exceeds 128, the oldest photos will be overwritten.
Command Reply	262,<result> result: Set the result of the command reply; IDn:OK/IDn:Err-The command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Examples	262,1,2 1# RS232 and 2# RS232 cameras take photos at the same time.
Remarks	

263 – Read the list of photos

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	263

Command Description	01 Get all the photo names of the device. 02 The device can upload up to 60 photo names in one package of data. If it is exceeded, the data will be uploaded in multiple packages. 03 VT200 Series devices support up to 128 photos storage.
Command Reply	263,<data> data: command reply; None-no photos; Unsupport-the instruction is not supported; Failed-the command format or password is wrong, the command failed; pt1,pt2,pt3,...ptn-photo names, separated by commas;
Examples	263
Remarks	

264 – Delete photo

Supported Modes	GPRS
Supported Models	VT200 Series, VT800-L
Command to Tracker	264,pt1,pt2,pt3,...ptn
Command Description	01 ptn: The name of the photo, which must be the same as the name uploaded in the photo event. 02 One command can delete max 60 photos.
Command Reply	264,<data> data: command reply; OK-The command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Examples	264,pt1,pt2 Delete pt1,pt2
Remarks	

265– Upload Photo

Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	265,photo_name
Command Description	01 photo_name: The name of the photo, which must be the same as the name uploaded in the photo event 02 The device will upload the photo data continuously in sub-packages until the complete photo data is uploaded, and each package uploads 1024 bytes of data. 03 The device does not automatically delete the photo after uploading the photo data, and requires the platform to issue a 264 command to delete it.

Command Reply	265,<data> data: command reply; photo_name,offset,size,data-photo data package; Unsupport-the command is not supported; Failed-The command format or password is wrong, or the photo does not exist. offset: the offset position of the photo data package in the entire photo data; size: The current data length of the data packet.
Examples	265,220516082549C1E34.jpg Read the photo named 220516082549C1E34.jpg
Remarks	

270-Set vehicle Type	
Supported Modes	GPRS/SMS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	270,car_type
Command Description	01 car_type: a 4-digit hexadecimal string, the default car type is 0x0000. 02 The OBDII protocol may be different for each vehicle. If failed to read data from C12 OBD Reader, the corresponding vehicle code needs to be set. 03 When when it is used in the car with the J1939 protocol, no setting is required. 04 The list of vehicle type codes are as follows: 0x0000 standard car series (default) 0x0100 Volkswagen/Audi/Skoda 0x0200 Buick/Chevrolet/Cadillac 0x0300 Ford/Mazda 0x0400 Toyota/Lexus 0x0500 Honda/Acura 0x0600 Nissan/Infiniti 0x0700 Kia 0x0800 modern 0x0900 BMW 0x0A00 Mercedes 0x0B00 Subaru 0x0C00 Mitsubishi 0x0D00 Renault 0x0E00 Peugeot/Citroen 0x0F00 Land Rover/Jaguar 0x1000 Volvo 0x1100 Ssangyong 0x1200 Chrysler 0x1300 Chery 0x1400 Geely 0x1500 BYD 0x1600 Great Wall 0x1700 seahorse 0x1800 China 0x1900 Trumpchi 0x1A00 MG 0x1B00 Qoros 0x1C00 Changan Automobile

	0x1D00 BAIC
Command Reply	270,<result> result: Set the result of the command reply; OK-the command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Examples	270,0400 Set model to 0400 (Toyota)
Remarks	

271 – OBD Command Transparent Transmission

Supported Mode	GPRS
Supported Models	VT200 Series, VT300-L, VT800-L
Command to Tracker	271
Command Description	01 The platform can send command to read OBD parameters or vehicle data according to the commands supported by OBD decoder 02 The device transmits data to the platform through transparent transmission
Command Reply	01 If the command is unsuccessful, reply: 271,<result> result: the result of Command Reply; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed. 02 If the command is successful, reply: 271,OBd data
Command Example	271
Remarks	

280 –Set WiFi AP

Supported Mode	GPRS/SMS
Supported Models	VT200 Series, VT800-L
Command to Tracker	280,index,ssid,password
Command Description	01 index: The location parameter number of the AP, supporting up to 32 APs; 02 ssid: SSID of AP, up to 32 characters; 03 password: AP login password, up to 32 characters. 04 When ssid and password are empty, delete the AP of index; 05 When index, ssid, and password are all empty, delete all APs. 06 When the WIFI network is normal, use WIFI to upload data first.
Command Reply	280, <result> result: Set the result of the command reply; OK-the command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.

Command Example	280,1,istartek,12345678 Set the SSID and password of the first AP.
Remarks	

282 – Set working mode (long standby device)

Supported Mode	GPRS/SMS
Supported Models	PT60-L
Command to Tracker	282,mode 282,3,interval_time 282,4,alarm_time1,alarm_time2,alarm_time3,...,alarm_timeN
Command Description	01 mode=0 is the normal working mode, device upload location data as per the parameter set by 102 command. 02 mode=1 is the normal mode of sports. In the moving (vibrating) state, the device uploads location data according to the normal_time of 102 command. After the device does not move for 3 minutes, it enters the sleep state (turn off the GPS power supply, GSM module sleep, MCU sleep), and upload the heartbeat packet data according to the time of 122 command. In the sleep state, you can receive SMS commands and incoming calls. 03 mode=2 is the sports power saving mode. In the moving (vibrating) state, the device uploads the location data according to the normal_time of the 102 command. After the device does not move for 3 minutes, it enters the sleep state (turn off the GPS power, turn off the GSM power, MCU sleep), upload the heartbeat packet data according to the time of the 122 command. SMS commands and incoming calls cannot be received in sleep mode. 04 mode=3 is the timing wake-up mode, the interval_time is in minutes, the maximum is 65535 minutes. The device wakes up regularly according to the interval_time to upload positioning data, and enters sleep mode after sending the data (turn off the GPS power, turn off the GSM power, and MCU sleep). In the sleep state, it cannot receive SMS commands and incoming calls, and the device cannot be woken up by movement (vibration). 05 mode=4 is the alarm clock mode, up to 24 alarm clocks can be set, the alarm clock parameter format is hh:mm, if the local time zone is set with the 601 command first, the alarm clock time is the local time, if no time zone is set, the alarm clock time is the time of GMT+0. The device wakes up according to the alarm clock time to upload location data, and enters sleep after sending the data (turn off the GPS power, turn off the GSM power, and MCU sleep). In the sleep state, it cannot receive SMS commands and incoming calls, and the device cannot be woken up by movement (vibration).
Command Reply	282,<result> result: Set the result of the command reply; OK-the command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Command Example	282,0 282,1 282,2 282,3,60 282,4,8:30,10:00,12:05,14:35,18:45,21:25,23:55
Remarks	

283 – Set Positioning Time after Waking Up from Power Saving Mode (long standby device)	
Supported Mode	GPRS/SMS
Supported Models	PT60
Command to Tracker	283,time
Command Description	01 time: The longest GPS positioning time after the device wakes up in power saving mode. After waking up, the device will get GPS position before generating positioning data. If the device fails to get position in this time, positioning data will be generated immediately; 02 The unit is seconds, the value range is 0~255 seconds, and 0 means no need to wait for GPS positioning. The default value is 60 seconds.
Command Reply	283,<result> result: Set the result of the command reply; OK-the command is set successfully; Unsupport-the command is not supported; Failed-The command format or password is incorrect, and the setting has failed.
Command Example	283,200 Set the positioning time after waking up from power saving mode to 200S
Remarks	

300 –Adding Bluetooth Peripherals	
Supported Mode	GPRS,SMS
Supported Models	VT300-L
Command to Tracker	300,<Index>,<Sensor Type>,<Name>,<MAC>
Command Description	Used to add different Bluetooth peripherals. Before using other Bluetooth-related commands, be sure to configure the Bluetooth index type in advance. <Index>: Range: 1-12 <Sensor Type>: Range: 0-11 0 - Unconfigured 1 - Temperature and Humidity Sensor (BT04/BT06) 2 - Door Magnetic Sensor 3 - Bluetooth Relay 4 - Bluetooth Ultrasonic Fuel Sensor (UL212) 5 - Bluetooth Heart Rate Sensor 6 - Bluetooth Speed Sensor 7 - Bluetooth capacitive fuel sensor (CPM212) 8 - Bluetooth Alcohol Sensor 9 - Bluetooth capacitive fuel sensor (Escort TD BLE) 10 - Temperature and Humidity Sensor (Escort TH BLE) 11 - Tire Pressure Sensor <Name>: Range: A-Z, a-z, 0-9, Length: 6 characters Customized name for the Bluetooth peripheral.

	<MAC> : Range: A-F, a-f, 0-9, Length: 12 characters MAC address of the Bluetooth peripheral
Command Reply	300,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	300,1,1,TEMP1,AABBCCDDEEFF Add a temperature and humidity sensor with MAC address AABBCCDDEEFF, index 1, and name it TEMP1
Remarks	The maximum number of Bluetooth peripherals that can be added is 12. Each peripheral has a unique MAC address. If a configuration error occurs, please check if there is a duplicate MAC address in another index number.

301 –Delete Bluetooth Peripherals

Supported Mode	GPRS
Supported Models	VT300-L
Command to Tracker	301,<Index>
Command Description	Deletes the Bluetooth peripheral at the specified index. <Index>: Range: 1-12 If index is empty, all Bluetooth peripherals will be deleted.
Command Reply	301,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	301,1 Delete the Bluetooth sensor at index 1. 301 Delete all Bluetooth peripherals.
Remarks	When the deleted Bluetooth peripheral is a temperature and humidity sensor, the temperature alarm threshold saved in the corresponding index will be deleted simultaneously

302 –Query Configured Bluetooth Peripherals

Supported Mode	GPRS
Supported Models	VT300-L
Command to Tracker	302,<Type>
Command Description	<Type>: Refer to the 300 instruction <Type> type table
Command	302,1,<Name1>,<MAC1>,2,<Name2>,<MAC2>..... 12,<Name12>,<MAC12>

Reply	Returns the Bluetooth configuration of the current Type
Command Example	302,1 Reply: 300,1,TEMP1,AABBCCDDEEFF,3,TEMP2,112233445566 This means that under the current index 1, there is a Bluetooth device named AABBCCDDEEFF with the name TEMP1, and under the index 3, there is a Bluetooth device named 112233445566 with the name TEMP2.
Remarks	

303 –Set the device Bluetooth Connection Password

Supported Mode	GPRS
Supported Models	VT300-L
Command to Tracker	303,<PWD>
Command Description	<PWD>: Four-digit password, range: A-Z, a-z, 0-9
Command Reply	303<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	303,8888 Reply: 300,OK Configure the Bluetooth connection password to 8888
Remarks	The Bluetooth connection password is empty by default. If the Bluetooth password is not configured, when you use the platform app to connect to the Tracker, the internal default password will be used to connect and ensure a successful connection.

600 – Set SMS Command Password

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L ,VT110 Series
Command to Tracker	600,password
Command Description	01 password: SMS command password, 4 characters, can be composed of any numbers, letters, and symbols; the default is 0000.
Command Reply	600,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	600,PW37 Set SMS command password as PW37

Remarks	
---------	--

601 – Set SMS Time Zone	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	601,time
Command Description	01 time: Time zone, the unit is minutes, the value range is -720~720, the default is UTC 0; 02 When the time zone is set, SMS data use time in the set time zone; 03 The time in GPRS data uses UTC0 time and cannot be set
Command Reply	601,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	601,480 Set SMS time zone to UTC+8 time zone 601,-480 Set the SMS time zone to UTC-8 time zone
Remarks	

602 – Set SIM Card PIN Code	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	602,pin
Command Description	01 pin: The PIN code of the SIM card, a 4-digit number;
Command Reply	602,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	602,1234 Set the PIN code to 1234
Remarks	

610 – Set Initial Mileage	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series

Command to Tracker	610,mileage
Command Description	01 mileage: The initial mileage in GPRS data, in meters.
Command Reply	610,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	610,1000 Set the initial mileage of GPRS data to 1000 meters.
Remarks	

611 – Clear GPRS/SMS Buffer

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	611,flag
Command Description	01 flag: =1: Clear the GPRS cache of server 1; =2: Clear the GPRS cache of server 2; =3: Clear SMS cache
Command Reply	611,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	611,1 Clear the GPRS cache of server 1.
Remarks	

612 – Initialization Parameter

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	612,flag
Command Description	01 flag: =1: Initialize other parameters except GPRS parameters such as IP, Port, APN, GPRS interval; =2: Initialize the SMS password to 0000; =3: Initialize all parameters.
Command Reply	612,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.

Command Example	612,3 Initialize all parameters
Remarks	

620 – Set Device LED Indicator Mode	
Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT200 Series
Command to Tracker	620,flag
Command Description	01 flag=0, when the device is working, the LED indicator works normally to indicate the working status. The default is 0; 02 flag=1, when the device is working, the LED is off without indication. 03 When the device is turned on, the LED indicator will work for 10 seconds and then turn off. When any button is pressed, the LED indicator will work for 10 seconds and then turn off.
Command Reply	620,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	620,1 Turn off the LED indication.
Remarks	

716 – Set Ignition Mode	
Supported Modes	GPRS/SMS
Supported Models	VT300-L
Command to Tracker	716,<Mode>,[<Volt>,<Motion Threshold>,<ACC_ON_TIME>,<ACC_OFF_TIME>]
Command Description	<Mode>: Ignition On/Off Mode Selection: Default: 0 0: Ignition off by ACC hardware detection 1: Ignition off by G-sensor detection 2: Ignition off by VIN voltage detection <Volt>: When <Mode> is 2, this value sets the voltage threshold for determining ignition on, accurate to one decimal place. Unit: V, Range: 7-90V, Default: 13.7V <Motion Threshold>: When <Mode> is 1, this value sets the vibration threshold for determining ignition on. Unit: mg, Range: 30-2000mg, Default: 65mg <ACC_ON_TIME>: When <Mode> is in non-ACC detection mode, ACC is considered on if the current voltage or continuous vibration lasts for the set time. Unit: seconds, Default: 5s. Range: 5 – 65535 <ACC_OFF_TIME>:

	When <Mode> is in non-ACC detection mode, if the current voltage or the duration of inactivity reaches the set time, the ACC is considered OFF. Unit: seconds, Default: 5 seconds. Range: 5 – 65535 All parameters following <Mode> can be omitted (e.g., 716, 1). If omitted, subsequent parameters will be initialized to default values based on the Mode value.
Command Reply	716,<result> result: the result of Command Reply; OK – The command is set successfully; Unsupport – The command is not supported; Failed – The command format or password is incorrect and the setting failed.
Command Example	716,0 Indicates the set point ignition mode is hardware ACC line ignition. 716,1 Indicates the set point ignition mode is vibration ignition and initializes subsequent parameters. The final result is: if the resultant vibration force is greater than 65mg for more than 5 seconds, it is considered ignition; if it is less than 65mg for more than 5 seconds, it is considered an ignition off. 716,2,13.7,,5,5 Indicates the set point ignition mode is voltage ignition. If the voltage is greater than 13.7V for more than 5 seconds, it is considered ignition; if the voltage is less than 13.7V for more than 5 seconds, it is considered an ignition off. 716,1,,65,5,60 Indicates the set point ignition mode is vibration ignition. If the resultant vibration force is greater than 65mg for more than 5 seconds, it is considered ignition; if it is less than 65mg for more than 60 seconds, it is considered an ignition off.
Remarks	

717 – Driving Behavior Direction Calibration	
Supported Modes	GPRS/SMS
Supported Models	VT300-L
Command to Tracker	717,<Mode>,[<Forward Direction X>,<Right Direction Y>,<Gravity Direction Z>]
Command Description	This command is used to calibrate the vehicle coordinates when the 718 command setting mode is set to G-sensor. <Mode>: Device vehicle coordinate system calibration mode: Default: 0 0: Automatic calibration: You must first secure the device to the vehicle, then issue the 717 command to initiate self-calibration. Then, start the vehicle and perform a quick acceleration by pressing the accelerator. Self-calibration will complete after approximately 2 seconds. 1: Manual calibration: Set the 717 parameters according to the planned forward, left, right, and gravity directions. Secure the device to the vehicle in the desired orientation. <Forward direction X>: When <Mode> is 1, this parameter sets the vehicle's forward direction. Default: X. Parameter range: {X, -X, Y, -Y, Z, -Z} <Right direction Y>: When <Mode> is 1, this parameter sets the vehicle's right-hand direction. Default: -Y.

	Parameter range: {X, -X, Y, -Y, Z, -Z} <Gravity direction Z>: When <Mode> is 1, this parameter sets the vehicle's gravity direction. Default: -Z. Parameter range: {X, -X, Y, -Y, Z, -Z}
Command Reply	717,<result> result: the result of Command Reply; OK – The command is set successfully; Unsupport – The command is not supported; Failed – The command format or password is incorrect and the setting failed.
Command Example	717,0 Set to automatic calibration mode 717,1,X,Y,Z Set the vehicle's forward direction to the X-axis, the right-hand direction to the Y-axis, and the gravity direction to the Z-axis. 717,1,X,-Y,-Z Set the vehicle's forward direction to the X-axis, the right-hand direction to the -Y-axis, and the gravity direction to the -Z-axis.  Vehicle coordinate system
Remarks	

718 – Driving Behavior Detection Mode Configuration	
Supported Modes	GPRS/SMS
Supported Models	VT300-L
Command to Tracker	718,<Mode>
Command Description	<Mode>: Device Vehicle Coordinate System Calibration Mode: Default: 0 0: Use GNSS data for driving behavior assessment. 1: Use the G-sensor accelerometer for driving behavior assessment. (Note: After setting 718, driving behavior calibration must be performed in conjunction with command 717.)
Command Reply	718,<result> result: the result of Command Reply; OK – The command is set successfully; Unsupport – The command is not supported; Failed – The command format or password is incorrect and the setting failed.

Command Example	718,1 Set to G-sensor acceleration sensor for driving behavior judgment.
Remarks	

750 – Log Upload	
Supported Modes	GPRS/SMS
Supported Models	VT300-L
Command to Tracker	750,sw,ftp_IP,ftp_port,ftp_login_name,ftp_login_password
Command Description	sw: 0 disables the automatic log upload function. 1 enables the automatic log upload function, uploading the log file to the FTP server with each reboot. 2 uploads the log file immediately without changing the original switch status. Note: "ftp_IP,ftp_port,ftp_login_name,ftp_login_password" is only required when sw is 2. ftp_IP: FTP server address, up to 32 characters. If left blank, the default is 218.17.189.150. ftp_port: FTP server port, up to 5 characters. If left blank, the default is 21. ftp_login_name: FTP server login name, up to 32 characters. If left blank, the default is test. ftp_login_password: FTP server login name, up to 32 characters. If left blank, the default is test. sw defaults to 0. The log file name format is "device number_time".txt, such as 862771072953238_20240924142201.TXT. The log upload path is "/t1log/".
Command Reply	750,<result> result: the result of Command Reply; OK – The command is set successfully; Unsupport – The command is not supported; Failed – The command format or password is incorrect and the setting failed. FTP busy! – There is currently a task in FTP, try again later.
Command Example	750,1 Enable active log reporting 750,2 Immediately report log files
Remarks	

751 – Static Drift Filter Settings	
Supported Modes	GPRS/SMS
Supported Models	VT300-L
Command to Tracker	751,sw
Command Description	sw: 0 disables static drift filtering. 1 enables static drift filtering, eliminating filtering for fixed-distance reports. This is enabled by default.
Command Reply	751,<result> result: the result of Command Reply;

	OK – The command is set successfully; Unsupport – The command is not supported; Failed – The command format or password is incorrect and the setting failed.
Command Example	751,1 Enable static drift filtering
Remarks	

800 – Get Current Location	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	800
Command Description	01 Get current location information; 02 When SMS Command to Tracker, reply the google map link of the location; 03 When GPRS Command to Tracker, reply the location data of 0 event.
Command Reply	01 If the command failed, reply: 800,<result> result: the result of Command Reply; Unsupport-The instruction is not supported; Failed-The command format or password is incorrect and the setting failed. 02 If the command is successful, the current location information will be returned.
Command Example	800
Remarks	

801 – Get Device SN, IMEI, Software Version	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	801
Command Description	01 Get the SN, IMEI, and software version of the device
Command Reply	01 If the command failed, reply: 801,<result> result: the result of Command Reply; Unsupport-The instruction is not supported; Failed-The command format or password is incorrect and the setting failed. 02 If the command is successful, reply: 801, SN, IMEI, software version
Command Example	801 801,143190870823,869912033012424,VT100_V100
Remarks	

802 – Get Device Working Status	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	802
Command Description	01 Get the working status of device, including device system status, input status, output status, CSQ value, satellite number, internal battery voltage, external power supply voltage
Command Reply	01 If the command failed, reply: 802,<result> result: the result of Command Reply; Unsupport-The instruction is not supported; Failed-The command format or password is incorrect and the setting failed. 02 If the command is successful, reply: 802, device system status, input status, output status, CSQ value, satellite number, internal battery voltage, external power supply voltage 03 Device system status, 8-digit hexadecimal character format, see the "system-sta" note in GPRS data. 04 Input status, 2-digit hexadecimal character format, see "in-sta" note in GPRS data. 05 Output status, 2-digit hexadecimal character format, see "out-sta" note in GPRS data. 06 CSQ value: decimal string. 07 satellite number: decimal string. 08 Internal battery voltage: decimal string, accurate to two decimal places. External power supply voltage: decimal string, accurate to two decimal places.
Command Example	802 802,0000000F,03,01,31,12,4.15,00.00
Remarks	

803 – Check SIM Card CCID	
Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	803
Command Description	Check SIM Card CCID
Command Reply	01 If the command is failed, reply: 803,<result> result: the result of the command reply; Unsupport-This parameter query is not supported; Failed-The command format or password is incorrect and the setting failed. 02 If the command is successful, reply: 803,CCID
Command Example	803 803,12345678901234567890
Remarks	

804 – Check SIM Card Balance

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT200 Series, PT60, VT300-L, VT800-L, VT120-L,VT110 Series
Command to Tracker	804,ussd_command
Command Description	01 ussd_command is the data service command
Command Reply	01 If the command is failed, reply: 804,<result> result: the result of the command reply; Unsupport-This parameter query is not supported; Failed-The command format or password is incorrect and the setting failed. 02 If the command is successful, reply: 804,string
Command Example	804, *121#
Remarks	This feature requires operator support

808 – Check device Parameters

Supported Modes	GPRS/SMS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	808,cmd
Command Description	01 Query the parameters of the device 02 cmd is the corresponding setting parameter command, such as 100, 102, please refer to the corresponding command description for the parameter content
Command Reply	01 If the command is failed, reply: 808,cmd,<result> result: the result of the command reply; Unsupport-This parameter query is not supported; 02 If the command is successful, reply: 808,cmd,parameters
Command	808,100
Example	808,100,1,device.istarmap.com,9094
Remarks	

900 – Output Control

Supported Modes	GPRS/SMS
-----------------	----------

Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	900,output,flag,time,safe_speed
Command Description	01 output: the serial number of the output port, 1= output1, 2= output2, 3= output3, and so on; 02 flag=1, turn on the output (the output is in low level state). flag=0, turn off the output (the output port is in a floating state) mode; 03 time: The time when the output is generated, the unit is ms, the value range is 0~65535; time=0 or empty, it is an infinite time output; 04 safe_speed: The speed limit of the action generated by the output, the unit is km/h, and the value range is 0~255; When safe_speed=0 or empty, the output immediately generates output action; When safe_speed>0, the output action is only generated when the current GPS speed is less than safe_speed and in the valid positioning state 05 The command control output has priority over the output generated by other events.
Command Reply	900,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	900,2,1 output2 produces output 900,1,1,100 Output1 generates 100ms output 900,1,1,0,10 Output1 produces output when the speed is lower than 10km/h
Remarks	

901 – Output Control (Mode Output)

Supported Modes	GPRS/SMS
Supported Models	VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	901,output,flag
Command Description	01 output: the serial number of the output port, 1= output1, 2= output2, 3= output3, and so on; 02 flag=1, the output action will be generated according to the mode set by the 251 command. flag=0, the output action will be stopped. 03 The command control output has priority over the output generated by other events.
Command Reply	901,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	901,2,1 Output2 produces output
Remarks	

910 – Restart Device or Module

Supported Modes	GPRS/SMS
Supported Models	PT60,VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L,VT110 Series
Command to Tracker	910,operation
Command Description	01 operation: =1: Restart the GSM module =2: Restart the GPS module =3: Restart the device
Command Reply	910,<result> result: the result of Command Reply; OK-The command is set successfully; Unsupport -the command is not supported; Failed-The command format or password is incorrect and the setting failed.
Command Example	910,3 Reboot the device
Remarks	

990 – FOTA Upgrade Detection

Supported Modes	GPRS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L
Command to Tracker	990,firmware_name
Command Description	01 firmware_name: The file name of the software package, ASCII character type, only the file name, without the directory. 02 After receiving the file name, the device compares with its own software version, and the reply sign informs the platform whether the upgrade can be performed. 03 FOTA starts with this command.
Command Reply	990,<flag><device code><maximum support FOTA single data packet length><software version> 01 flag: comparison result of software file name, 1 byte, hexadecimal data type. =0x01, indicating that the OTA file matches the version of the device itself, and the upgrade can be performed. =0x02, it means that the OTA file does not match the version of the device itself. If upgrade may cause the device to crash, the platform should pop up a prompt window. If you are sure to upgrade, you can directly perform the upgrade. 02 Machine code: 2 bytes, hexadecimal character type, big-endian alignment. 03 Maximum support for FOTA single data packet length: the data length of a single software package, 2 bytes, hexadecimal character type, big-endian alignment, the default support is 1024 bytes (0x0400). Software version: ASCII character type, the software version of the device itself.
Command Example	Send: 990,VT100_V101 Answer: 990,<0x01><0x0003><0x0400><VT100_V100>
Remarks	

991 – FOTA Upgrade Authorization

Supported Modes	GPRS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L
Command to Tracker	991
Command Description	01 The platform issues instructions to authorize the device to enter the OTA state. After OTA authorization, if the device fails to receive the OTA data packet from the platform for more than 20 minutes, it will exit the OTA state. At this time, the platform needs to perform the 990->991 process again. 02 After OTA is authorized, the device will no longer send timing interval data or alarm data until it exits the OTA state.
Command Reply	991,<flag> flag: 1 byte, hexadecimal data type. =0x01 means authorization is successful. =0x00 means authorization failed. =0xFF, OTA authorization is illegal, that is, the 990 command is not issued in advance to upgrade the detection.
Command Example	Send: 991 Answer: 991,<0x01>
Remarks	

992 – FOTA Data packet delivery

Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L
Command to Tracker	992, <OTA packet address offset (4 bytes)> <OTA packet length (2 bytes)> <OTA packet>
Command Description	01 OTA data packet address offset: the address offset of the OTA data packet delivered this time in the entire OTA file, 4 bytes are big-endian aligned, hexadecimal number; 02 OTA data packet length: the length of the OTA data packet sent this time, 2 bytes are big endian, hexadecimal number; 03 OTA data packet: The n-byte OTA data packet sent this time (assuming the length of the OTA data packet is n).
Command Reply	992, <OTA packet address offset (4 bytes)> <OTA packet length (2 bytes)> <flag (1 byte)> 01 OTA data packet address offset: corresponding to the data packet address offset sent from the platform; 02 OTA data packet length: corresponding to the data packet length sent by the platform; 03 flag: = 0x01, received and saved correctly, the platform can then send the OTA data packet; = 0x00, there is an error in receiving or saving, and the platform needs to resend the same OTA data packet to the terminal; = 0xFF, OTA is not authorized or authorization timed out.
Command Example	Send: 992,<00 00 00 00 04 00 1A 68 92 C2 93 F6 D8 90 71 08 3A C2 F1 5E 80 88 3D 58 0A 12 85 F6 D8 90 69 08 3A C2 DA 6A 7A 5C 02 D2 22 F2 42 12 62 32 82 52 A2 72 F1 E6 48 80 FD A0 92 EA 42 12 62 32 B5 EE 90 18 19 80 72 CA 19 56 38 F0 83 CC 9E C6 45 A2 64 0C 0F 78 2A 32 F1 2E D0 58 5B A4 36 2E 57 45 E2 BA A9 A6 08 40 B3 BC 0E B6 D5 B2 F4 1C FF E8 1A A2 81 3E 60 68 CB 54 66 DE

	6D 9C 90 18 1B 64 F6 EE DD FA 7C D4 07 30 A2 7A 69 66 C8 00 73 7C CE 76 95 72 B4 DC BF A8 DA 62 41 FE 20 28 8B 14 26 9E 2D 8A 8C 64 77 20 12 6A F9 76 58 10 E3 6C 3E 66 25 42 04 AC DD 86 46 BE DF DF A6 1E AD 0A 0C E4 F7 A0 92 EA 79 F6 D8 90 63 EC BE E6 A5 C2 84 2C 6F 98 4A 52 D1 CE 70 F8 99 E1 E5 5E E7 BC 21 FA 8B E0 E5 5E ED 4A 4C 24 37 E0 D2 2A B9 36 18 D0 A3 2C FE 26 E5 02 C4 6C AF D8 8A 92 11 0E B0 38 7B 84 16 0E 3D 1A 9C F4 67 50 C2 9A 49 06 68 A0 D3 9C EE 96 83 33 87 C0 33 3C 8E 36 55 32 74 9C 7F 68 9A 22 01 BE E0 E8 4B D4 E6 5E ED 4A 4C 24 37 E0 D2 2A 82 A2 E0 4A 12 92 36 D2 D0 CA EA E2 DA 66 40 D6 29 C8 46 81 C3 63 A0 4F B5 77 88 38 31 62 45 E6 91 24 38 AA ED 5C AF 62 A3 4B F2 A5 FF 90 F9 6C BD ED 68 C7 FE 1A 3F 40 20 F4 A9 BA 40 1E 7D 5C 2C 00 FC 14 ED A5 F3 86 61 AC 38 FF FE 46 6F BB 6B D9 F2 35 04 CC 65 8D 4C 2E 29 0E 1E 26 4F BB F2 58 67 50 89 F1 45 BF 03 F6 5E D9 36 CB 8A D2 DB 77 3F D0 C0 B7 2B 85 2B 46 5C 6B 3B 64 BD 25 8A F1 33 18 AC 38 B1 69 F5 1E 07 23 53 DE A5 1B E1 80 44 79 5C EB 82 52 A2 51 94 90 D4 D1 E6 70 96 2C FE 29 50 C9 61 64 CE 27 F6 F1 00 71 CA DD F7 B2 12 15 D8 B5 0E 4E 11 90 E8 49 C2 62 1A A2 F2 62 22 A5 D3 F7 9A C0 7D 26 ED 51 AE 5B 80 AD 7A FA BC 28 E3 52 A7 E1 6D CD B9 8E 69 4D 5A 1A 7E 48 11 80 AB 4D 16 B7 A5 54 85 DA DF 78 94 B4 DD 1E 56 81 01 52 2A 7D 9A BF E8 B1 A6 A3 64 D5 E2 07 60 69 2E EB DC 8D 6A 4F D8 21 4C 98 E5 B3 59 C4 B8 A5 00 B7 2E DE BB 3F BE E6 AF C1 B2 0A 2A 96 FA AA 92 30 9A 0A 72 1C C2 B2 6A 54 BA 6A EA AD 14 34 5B CE 56 A3 DC 79 DD 4A 16 40 98 A8 FF EF 38 34 2D 62 FE 99 90 D0 F4 34 59 1F B8 A9 FE 94 64 84 DA 4A 70 EA 3B 14 FC 3A 08 57 F6 6D 05 12 CC F2 46 43 0D 9F 50 A7 BE 64 3E B2 5E CC 2C A9 F1 4A 22 54 2A 1D 6C 53 2F 54 6C F4 C1 27 99 6A 63 F4 7E 58 90 05 85 00 14 53 78 FC 3B 42 E8 0E 53 6A 25 33 B4 ED 8C 1D 7F A8 54 C8 AE 59 40 26 01 4E 7C 12 5F C1 DE 29 63 1F 5E B0 3F 3F B6 AC BB D0 88 B9 CA DA 47 7D 26 47 CF 80 36 03 78 A0 3D 2B 4D B0 F0 D2 77 18 87 D5 08 C8 88 E0 4B 24 EE AF A0 A0 B3 92 21 88 91 B2 20 BD 65 03 89 8E 80 B2 AB 13 AE D7 9C 28 AD 7B A1 92 6D 8F 52 22 72 E3 AE 5B A9 C1 26 5C 35 7C 0C E7 41 A6 5B 75 E2 14 04 64 C4 00 64 D2 91 76 E2 9A 67 6D 45 B6 7E E6 34 0A 6D D2 BA FB 9C 0C 80 04 1C 7C DC 3C 9C FC 71 CE 7E AE 75 A0 A0 A0 80 A0 A0 D0 F5 BE D4 A7 92 B9 2C D2 DA BD BF 02 52 03 AC 3D B3 7A 09 1E 94 74 DA 4C 1E 21 FA 9E 9C EE 6E A0 81 76 FB E8 EF 45 CC 8C 9C 63 C4 95 6E 08 C8 0E 19 38 D2 97 A9 2B 0D BE C6 6B B1 51 32 18 90 B8 7F F2 33 38 C5 40 C2 EB A5 57 9A AE 13 06 D0 BB D0 08 78 78 C8 DF 4E 58 98 77 38 FB 04 9E FC AE 8E AE 1E BD CC 98 A8 4D B2 93 BC D5 FD 44 F1 B0 0D 1E C4 D0 17 D7 D0 09 D1 FC A4 FD 15 E8 9F 9F C8 C1 A9 26 F9 FE D0 BA 1D 3C 64 BD 25 48 3F 2E 26 4F EB 87 A8 5B 8D 10 60 24 58 90 62 3C AC 42 35 0D 0A Answer: 992,<0x00000000> <0x0400> <0x01>
Remarks	Refer to Appendix B

993 – FOTA Data Check	
Supported Modes	GPRS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L
Command to Tracker	993, <OTA data length (4 bytes)> <OTA data check code (2 bytes)>
Command Description	01 After the platform sends all the OTA data to the device, it checks whether the data received by the device is wrong. 02 OTA data length: the length of all OTA data (excluding the checksum of the last two bytes of the OTA file), 4 bytes are big-endian aligned, hexadecimal number. 03 OTA data check code: the checksum of all OTA data, the last two bytes of OTA file, hexadecimal number.
Command Reply	993,<flag(1byte)> flag: = 0x01, the data verification is correct, the platform can send 994 command to update the firmware;

	= 0x00, data verification error, need to restart FOTA process; = 0xFF, OTA is not authorized or authorization timed out.
Command Example	Send: 993,<0x00017800><0x7E94> Answer: 993,<0x01>
Remarks	Refer to Appendix B

994 – Device Update FOTA Program	
Supported Modes	GPRS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L
Command to Tracker	994
Command Description	01 After getting the correct verification result of the terminal, the platform instructs the device to update the OTA program. Note: You must ensure that the checksum is correct before using this command, otherwise the device may crash.
Command Reply	994,<flag> = 0x01, the device receives the command and is ready to upgrade, and reply to the command before the program is updated; = 0x02, the device upgrade is completed, and the command will be replied after the program update is completed; = 0x00, device upgrade failed; = 0xFF, OTA is not authorized or authorization timed out.

995 – Cancel FOTA	
Supported Modes	GPRS
Supported Models	PT60-L, VT100 Series, VT120-L, VT200 Series, VT300-L, VT800-L
Command to Tracker	995
Command Description	After FOTA is authorized, the platform can cancel FOTA accordingly.
Command Reply	995,<flag> = 0x01, FOTA cancel success.
Command Example	Send: 995 Answer: 995,<0x01>
Remarks	

Appendix A-Alarm Event Codes and Alarm Characters

The following is the corresponding table of alm-code and alm-data:

alm-code	alm-data	SMS Alarm Head String	Description	Supported Models
0	NULL	Interval	Time interval tracking	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
1	NULL	SOS	Input1 active	VT100Series,VT200Series,VT300-L,VT800-L,VT110Series
2	NULL	IN1 Inactive	Input1 inactive	VT100Series,VT200Series,VT300-L,VT800-L,VT110Series
3	NULL	Ignition On	Input2 active	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
4	NULL	Ignition Off	Input2 inactive	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
5	NULL	Door Open	Input3 active or BLE Door peripheral Open(VT300-L)	VT200Series,VT300-L,VT800-L VT300-L: Index 0 indicates that digital input 3 is triggered. Index 1-12 indicates that the Bluetooth door sensor is triggered.
6	NULL	Door Close	Input3 inactive or BLE Door peripheral Close(VT300-L)	VT200Series,VT300-L,VT800-L VT300-L: Index 0 indicates that digital input 3 is not triggered. Index 1-12 indicates that the Bluetooth door sensor is not triggered.
7	NULL	IN4	Input4 active	VT200Series,VT800-L
8	NULL	IN4 Inactive	Input4 inactive	VT200Series,VT800-L
9	NULL	IN5	Input5 active	VT800-L
10	NULL	IN5 Inactive	Input5 inactive	VT800-L
11	NULL	IN6	Input6 active	
12	NULL	IN6 Inactive	Input6 inactive	
17	NULL	Low Ext-Power	Ext-power low	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
18	NULL	Ext-Power Cut	Ext-power lost	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
19	NULL	Ext-Power On	Ext-power re-connect	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
20	NULL	Low Battery	Internal battery low	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
21	NULL	GPS Antenna Cut	GPS antenna cut	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
22	NULL	Speeding	Speeding alarm	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
23	NULL	Enter Sleep	Enter sleep	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
24	Trigger	Wake Up	Exit sleep	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series

25	n: index of fence	Exit Fence	Exit geo-fence	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
26	n: index of fence	Enter Fence	Enter geo-fence	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
27	NULL	Lose GPS Signal	Lose GPS Signal	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
28	NULL	Get GPS Signal	Get GPS Signal	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
29	NULL	Heading Change	Heading Change	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
30	NULL	Distance	Distance tracking	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
31	NULL	/	Heartbeat Report	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
32	NULL	Power On	Device Power On	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
33	NULL	Power Off	Device Power Off	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L
34	NULL	Current	Reply Current	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
35	NULL	Tow	Tow Alarm	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
36	NULL	Stop Moving	Stop Moving	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
37	NULL	Start Moving	Start Moving	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
38	NULL	Idling	Idling Alarm	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
39	NULL	Harsh Accelerate	Harsh accelerate	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
40	NULL	Harsh Braking	Harsh braking	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
41	NULL	Harsh Turning	Harsh Turning	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
42	NULL	Impact	Impact Alarm	VT100Series,VT200Series,PT60,VT300-L,VT120-L,VT800-L,VT110Series
43	NULL	Fatigue Driving	Fatigue driving	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
44	NULL	Fatigue Relieve	Fatigue relieve	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
45	NULL	Overtime Driving	Overtime driving	VT100Series,VT200Series,VT300-L,VT120-L,VT800-L,VT110Series
46	n: index of Temperature	Temperature High	Temperature High	VT200Series,VT300-L,VT800-L Index: 1-8 for 1-wire sensors 11-22 for Bluetooth temperature sensors (offset index is Bluetooth index 1-index 12)

47	n: index of Temperature	Temperature Low	Temperature Low	VT200Series, VT300-L,VT800-L Index: 1-8 for 1-wire sensors 11-22 for Bluetooth temperature sensors (offset index is Bluetooth index 1-index 12)
48	n: index of tank	Fuel Steal	Fuel Steal Alarm	VT100Series,VT200Series, VT300-L,VT120-L,VT800-L
49	n: index of tank	Fuel Level Low	Low Fuel Alarm	VT100Series,VT200Series, VT300-L,VT120-L,VT800-L
50	NULL	GSM Jammed	GSM Jammed	VT100Series,VT200Series,PT60, VT300-L,VT120-L,VT800-L,VT110Series
51	NULL	GSM No Jamming	GSM No Jamming	VT100Series,VT200Series,PT60, VT300-L,VT120-L,VT800-L,VT110Series
52	Trigger	Stealing	Stealing Alarm	VT100Series,VT200Series,PT60, VT300-L,VT120-L,VT800-L,VT110Series
53	RFID/iButton number, Hexadecimal character	Swiping Card	RFID Event	VT200Series, VT300-L,VT800-L
54	Picture name, character	/	Picture Event	VT200Series,VT800-L
55	NULL	dismantling	Anti Disassembly Alarm	PT60
56	Trouble Code, Hexadecimal character	Trouble	Diagnostic Trouble Code (diagnostic trouble code alarm). The first byte indicates the fault category, 00 is J1979 fault, 01 is J1939 fault; every subsequent 3 bytes indicate a fault code. J1979 fault code: The first two bytes indicate the fault code, and the last byte indicates the status (01: indicates the solved fault, 02: represents the pending fault). For example: 00019501019602 It is J1979 fault type, fault code is P0195, P0196. J1939 fault code: 3 bytes are represented as 19 bits	VT200Series, VT300-L,VT800-L

			(SPN) + 5 bits (FMI) For example: 01F40305F8 0307 It is J1939 fault type, fault code is SPN1012-5, SPN1016-7	
57	Low Speed	Low Speed	Low Speed Alarm	VT100Series,VT200Series,PT60 VT300-L,VT120-L,VT800-L,VT110Series
58	Sign out	Sign out	sign out Alar	VT200Series,VT300-L,VT800-L

Appendix B – Example FOTA process

The FOTA process must follow the following instruction steps:

990 -> 991 -> 992 -> 993 -> 994

Examples of FOTA file data are as follows:1A 68 92 C2 93 F6 D8 90 71 08 3A C2 F1 5E 80 88 3D 58 0A
12 85 F6 D8 90 69 08 3A C2 DA 6A 7A 5C 02 D2 22 F2 42 12 62 32 82 52 A2 72 F1 E6 48 80 FD A0 92
EA 42 12 62 32 B5 EE 90 18 19 80 72 CA 19 56 38 F0 83 CC 9E C6 45 A2 64 0C 0F 78 2A 32 F1 2E D0
58 5B A4 36 2E 57 45 E2 BA A9 A6 08 40 B3 BC 0E B6 D5 B2 F4 1C FF E8 1A A2 81 3E 60 68 CB 54 66
DE 6D 9C 90 18 1B 64 F6 EE DD FA 7C D4 07 30 A2 7A 69 66 C8 00 73 7C CE 76 95 72 B4 DC BF A8
DA 62 41 FE 20 28 8B 14 26 9E 2D 8A 8C 64 77 20 12 6A F9 76 58 10 E3 6C 3E 66 25 42 04 AC DD 86
46 BE DF DF A6 1E AD 0A 0C E4 F7 A0 92 EA 79 F6 D8 90 63 EC BE E6 A5 C2 84 2C 6F 98 4A 52 D1 CE
70 F8 99 E1 E5 5E E7 BC 21 FA 8B E0 E5 5E ED 4A 4C 24 37 E0 D2 2A B9 36 18 D0 A3 2C FE 26 E5 02
C4 6C AF D8 8A 92 11 0E B0 38 7B 84 16 0E 3D 1A 9C F4 67 50 C2 9A 49 06 68 A0 D3 9C EE 96 83 33
87 C0 33 3C 8E 36 55 32 74 9C 7F 68 9A 22 01 BE E0 E8 4B D4 E6 5E ED 4A 4C 24 37 E0 D2 2A 82 A2
E0 4A 12 92 36 D2 D0 CA EA E2 DA 66 40 D6 29 C8 46 81 C3 63 A0 4F B5 77 88 38 31 62 45 E6 91 24
38 AA ED 5C AF 62 A3 4B F2 A5 FF 90 F9 6C BD ED 68 C7 FE 1A 3F 40 20 F4 A9 BA 40 1E 7D 5C 2C 00
FC 14 ED A5 F3 86 61 AC 38 FF FE 46 6F BB 6B D9 F2 35 04 CC 65 8D 4C 2E 29 0E 1E 26 4F BB F2 58
67 50 89 F1 45 BF 03 F6 5E D9 36 CB 8A D2 DB 77 3F D0 C0 B7 2B 85 2B 46 5C 6B 3B 64 BD 25 8A F1
33 18 AC 38 B1 69 F5 1E 07 23 53 DE A5 1B E1 80 44 79 5C EB 82 52 A2 51 94 90 D4 D1 E6 70 96 2C
FE 29 50 C9 61 64 CE 27 F6 F1 00 71 CA DD F7 B2 12 15 D8 B5 0E 4E 11 90 E8 49 C2 62 1A A2 F2 62
22 A5 D3 F7 9A C0 7D 26 ED 51 AE 5B 80 AD 7A FA BC 28 E3 52 A7 E1 6D CD B9 8E 69 4D 5A 1A 7E
48 11 80 AB 4D 16 B7 A5 54 85 DA DF 78 94 B4 DD 1E 56 81 01 52 2A 7D 9A BF E8 B1 A6 A3 64 D5
E2 07 60 69 2E EB DC 8D 6A 4F D8 21 4C 98 E5 B3 59 C4 B8 A5 00 B7 2E DE BB 3F BE E6 AF C1 B2 0A
2A 96 FA AA 92 30 9A 0A 72 1C C2 B2 6A 54 BA 6A EA AD 14 34 5B CE 56 A3 DC 79 DD 4A 16 40 98
A8 FF EF 38 34 2D 62 FE 99 90 D0 F4 34 59 1F B8 A9 FE 94 64 84 DA 4A 70 EA 3B 14 FC 3A 08 57 F6
6D 05 12 CC F2 46 43 0D 9F 50 A7 BE 64 3E B2 5E CC 2C A9 F1 4A 22 54 2A 1D 6C 53 2F 54 6C F4 C1
27 99 6A 63 F4 7E 58 90 05 85 00 14 53 78 FC 3B 42 E8 0E 53 6A 25 33 B4 ED 8C 1D 7F A8 54 C8 AE
59 40 26 01 4E 7C 12 5F C1 DE 29 63 1F 5E B0 3F 3F B6 AC BB D0 88 B9 CA DA 47 7D 26 47 CF 80 36
03 78 A0 3D 2B 4D B0 F0 D2 77 18 87 D5 08 C8 88 E0 4B 24 EE AF A0 A0 B3 92 21 88 91 B2 20 BD 65
03 89 8E 80 B2 AB 13 AE D7 9C 28 AD 7B A1 92 6D 8F 52 22 72 E3 AE 5B A9 C1 26 5C 35 7C 0C E7 41
A6 5B 75 E2 14 04 64 C4 00 64 D2 91 76 E2 9A 67 6D 45 B6 7E E6 34 0A 6D D2 BA FB 9C 0C 80 04 1C
7C DC 3C 9C FC 71 CE 7E AE 75 A0 A0 A0 80 A0 A0 D0 F5 BE D4 A7 92 B9 2C D2 DA BD BF 02 52
03 AC 3D B3 7A 09 1E 94 74 DA 4C 1E 21 FA 9E 9C EE 6E A0 81 76 FB E8 EF 45 CC 8C 9C 63 C4 95 6E
08 C8 0E 19 38 D2 97 A9 2B 0D BE C6 6B B1 51 32 18 90 B8 7F F2 33 38 C5 40 C2 EB A5 57 9A AE 13



File Name	iStartek GPS Tracker Protocol	Ver.	2.3
Update	2025-10-18	Page	75

06 D0 BB D0 08 78 78 C8 DF 4E 58 98 77 38 FB 04 9E FC AE 8E AE 1E BD CC 98 A8 4D B2 93 BC D5 FD
44 F1 B0 0D 1E C4 D0 17 D7 D0 09 D1 FC A4 FD 15 E8 9F 9F C8 C1 A9 26 F9 FE D0 BA 1D 3C 64 BD
25 48 3F 2E 26 4F EB 87 A8 5B 8D 10 60 24 58 90 62 3C AC F4 DC ED 96 3A 55 18 1A 25 9E 82 3D E8
20 40 98 1D 85 5E ED 46 CD 3D C6 EA A5 A0 18 94 63 61 63 46 36 74 B1 5C 3F 3F 28 21 C9 B5 CE 37
33 66 E1 B1 6A 33 7F 9E 2D CC 54 2D E5 20 D8 51 69 1C BB 0A E2 6B 67 92 95 3E B0 5F 8D 53 6C 45
AA EE 5A E6 6A 8C 9C 6E 48 4F 3E DE 0C BA C7 B0 00 62 14 A4 24 37 C6 22 02 59 14 78 15 72 6A DA
77 AA 83 FA DD FC A4 FD 05 18 7F 4E E6 0C 95 61 39 C8 F5 E6 A6 8B E0 07 3C 57 A4 04 D7 86 0E 77
13 6E E6 0F BB 1E 69 68 50 89 B1 4C 5B 3D 9C 6C 4E D0 E5 39 6A A4 BE 32 4E 94 ED B4 79 58 50 F4
50 00 48 BC 58 59 EB 1C 9D 71 44 0F D9 C1 DD AC 4B BA 02 F6 82 6A BD D0 FC 41 36 DA E5 D9 32
AA 9D FC 64 B8 37 3A AD 3C CB 2A 7D BC C4 1E 0B 37 78 28 A7 FE B9 38 00 B9 61 1C 2B B2 B5 2C
BB A2 25 3C CB 72 E7 FC AC 55 D6 7E C6 CE B3 E0 C5 0E 24 7A FA 59 A4 F8 3C EC E1 E3 79 C2 45 61
58 CA 38 3C 38 37 16 BC ED 2A F6 2F D8 C8 D6 38 49 AE 57 E1 22 72 53 55 06 D6 26 77 18 50 FA 9E
7D 4A 11 59 89 10 70 D0 77 6F F9 35 77 37 9E A2 AF 23 0C 4D 75 60 60 C0 01 12 EA 62 F4 88 4E D2
04 F5 B7 BA CA 5A CC 4A 42 26 63 2D FF 3A 8C 22 68 38 4F A6 5C 76 BA B0 8A 36 24 B8 33 CC 1B C6
3E D6 80 FC A3 18 BE 0B B9 8C EC 4A 30 44 31 AB AD 4E 8E CE C9 B0 3A 04 F8 14 36 60 46 A3 AB FD
29 44 24 07 5E BA D7 C1 B3 EC 5C 71 66 46 81 28 E2 6C 90 AC CE 98 DE F9 3F B1 07 1F BE 0A 7C 3D
3F 82 D2 82 1A 52 A4 B8 9F 96 78 E5 9A A2 65 CD 5E 55 37 4E FE 29 FB C6 93 84 56 87 96 6A 8A 4D
4C E1 10 66 B0 E6 A6 D0 6E C6 77 27 96 09 78 98 F5 2A E4 4A F7 6F 6D 65 D6 D6 E3 78 62 3C AC F7
52 93 0C 0C 94 A4 5E F8 B0 40 48 E9 C2 A2 1A DA E2 C2 3A 25 70 F1 F9 B7 EC C4 4A 12 99 A9 02 E2
DA 9A 22 02 FA 1E 8D 34 E5 46 4E 03 28 80 50 4F B4 4C 02 8A E1 F1 EA B6 71 CC 14 E8 F7 8A 5D 0C
9B 60 85 44 F3 33 FC 16 E3 06 41 11 0A D3 DF 32 55 14 AC 46 B3 D6 F1 30 48 41 39 74 C3 1A E2 4A
0D 9C 77 DA 4A 7B B5 04 E2 2C BF 28 C8 30 E7 DE BB 7F 0E 23 99 53 45 66 F7 DD 23 D1 8F E1 AE 8F
55 6D E0 61 84 B2 EC 7D 2B F1 EA 27 96 B6 1B 98 58 FD 4D D3 DD FC 74 3C 6C 3C 1C 0C 9B B2 4A
92 95 D4 4F 72 E2 A7 3A 7F 59 29 0D 42 36 D9 0E D7 06 7F 64 9A BA BD 64 81 30 7E 36 76 33 D7 31
A0 55 1F D4 3C FC D3 02 42 14 E5 A4 B4 D3 25 64 74 9C 0B 1A 1A A1 AD 8E A9 31 2E 2C 6E 42 FA 43
80 E4 6C 05 12 06 5E 47 7C 80 68 61 F7 62 98 0F 42 F0 89 0B 2E 5E 2E B8 D1 A7 58 63 60 60 60 28 80
37 42 8D E0 C0 D0 B5 7A 29 29 3E 8A A2 E3 6C A4 92 F7 5B 57 1E 11 7E 23 E4 61 CB 5D 32 95 57 69
5A 3A C7 24 E4 94 D9 B6 E3 5B C8 BE CE 82 92 82 7F F0 AC C8 7A F3 6A 1B E7 22 2B BE 02 EF 59 0C
CE 75 EA 56 EE C7 58 78 86 0B 6F 12 18 86 AA 3E 82 93 49 FE C2 A4 94 04 84 74 F9 D6 03 74 C6 E9
D8 5E 3F 60 90 D9 7D DA A6 DC ED 9E A8 9E 3E 8A F1 73 48 67 7B FD 28 D1 F0 92 C2 C2 3F 2A BE
7C 02 D3 84 E4 A4 0D A9 88 4B B4 A0 FD 92 16 8C 74 7F 8B 48 84 D8 3D 6A AE 32 70 E3 C8 DB 77
5B 37 E0 1B A4 DD 2E 8E 93 D2 D4 64 49 3E B6 E6 77 D4 42 C7 9D 46 58 EA A6 40 10 90 F0 10 D8 A9
5E C3 DE 18 09 77 9F B2 B5 6C 49 AF 5A F4 87 BA 6B 64 6E 51 10 F3 30 59 D2 89 72 FD 2F BF 54 7F
51 E2 67 FD CE A3 C6 AC 59 BF 60 FB 40 AD 06 66 66 36 78 57 44 59 59 3A CC A3 2E C8 D8 E8 65 82
F7 87 12 D7 C4 0D 06 FB 88 25 9F A2 07 F7 6A 9C DC 2C 41 20 90 0F B4

.....

.....

BD E0 23 C8 41 F4 07 BC 75 80 FF 03 A2 6F EE 61 36 6F 16 12 A8 41 F4 53 B7 DE CB 22 3D 72 8B 93
E4 CB 20 DD D9 E6 54 DB 5C F4 EA AA E5 03 6F 86 98 37 90 80 56 15 D8 ED C7 68 C5 B0 29 10 D8 41
00 B1 AC 14 16 6A 5D 85 19 3C DD E4 D7 4E F7 DD 02 77 8E 3B 72 8B 90 78 ED 28 E1 88 8C 39 01 F9
B7 B6 B0 A8 5D 58 51 78 7F 65 35 72 AC E5 54 DE 09 17 49 CF 96 C3 3A 82 BB 0E F3 BE B6 92 23 2A
92 69 4C 2D B4 9B 8C B7 0E F4 C1 57 01 F6 42 84 8F CE 8D 88 81 28 2E F5 32 7F 4F D3 ED 82 36 78

File Name	iStartek GPS Tracker Protocol	Ver.	2.3
Update	2025-10-18	Page	76

[illegible]

Explanation: The above FOTA file is only an example of the head and tail data of the file. The last two bytes 0x7E 0x94 are the checksum of the FOTA file. The checksum does not need to be sent to the machine in the 992 command, only in the 993 command Send it to the device for checksum verification. The whole FOTA data length does not include the last two bytes 0x7E 0x94.

Step 1: Send 990 command for version upgrade detection:

Send: 24 24 61 34 36 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 30 2C 56 54 32 30 30 5F 4C 5F 56 31 31 39 5F 32 30 32 32 30 38 30 33 2E 42 69 6E 41 36 0D 0A

Upload: 26 26 7E 33 35 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 30 2C 01 00 21 04
00 56 54 32 30 30 5F 4C 5F 56 31 39 0D 0A

Step 2: Send 991 order FOTA authorization:

Send: 24 24 3A 32 30 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 31 44 45 0D 0A

Upload: 26 26 3A 32 32 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 31 2C 01 31 31

0D 0A

Step 3: Send 992 command to download FOTA data:

Send: 24 24 3B 31 30 35 31 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 32 2C 00 00
00 00 04 00 1A 68 92 C2 93 F6 D8 90 71 08 3A C2 F1 5E 80 88 3D 58 0A 12 85 F6 D8 90 69 08 3A C2
DA 6A 7A 5C 02 D2 22 F2 42 12 62 32 82 52 A2 72 F1 E6 48 80 FD A0 92 EA 42 12 62 32 B5 EE 90 18
19 80 72 CA 19 56 38 F0 83 CC 9E C6 45 A2 64 0C 0F 78 2A 32 F1 2E D0 58 5B A4 36 2E 57 45 E2 BA
A9 A6 08 40 B3 BC 0E B6 D5 B2 F4 1C FF E8 1A A2 81 3E 60 68 CB 54 66 DE 6D 9C 90 18 1B 64 F6 EE
DD FA 7C D4 07 30 A2 7A 69 66 C8 00 73 7C CE 76 95 72 B4 DC BF A8 DA 62 41 FE 20 28 8B 14 26 9E
2D 8A 8C 64 77 20 12 6A F9 76 58 10 E3 6C 3E 66 25 42 04 AC DD 86 46 BE DF DF A6 1E AD 0A 0C E4
F7 A0 92 EA 79 F6 D8 90 63 EC BE E6 A5 C2 84 2C 6F 98 4A 52 D1 CE 70 F8 99 E1 E5 5E E7 BC 21 FA
8B E0 E5 5E ED 4A 4C 24 37 E0 D2 2A B9 36 18 D0 A3 2C FE 26 E5 02 C4 6C AF D8 8A 92 11 0E B0 38
7B 84 16 0E 3D 1A 9C F4 67 50 C2 9A 49 06 68 A0 D3 9C EE 96 83 33 87 C0 33 3C 8E 36 55 32 74 9C
7F 68 9A 22 01 BE E0 E8 4B D4 E6 5E ED 4A 4C 24 37 E0 D2 2A 82 A2 E0 4A 12 92 36 D2 D0 CA EA E2
DA 66 40 D6 29 C8 46 81 C3 63 A0 4F B5 77 88 38 31 62 45 E6 91 24 38 AA ED 5C AF 62 A3 4B F2 A5
FF 90 F9 6C BD ED 68 C7 FE 1A 3F 40 20 F4 A9 BA 40 1E 7D 5C 2C 00 FC 14 ED A5 F3 86 61 AC 38 FF
FE 46 6F BB 6B D9 F2 35 04 CC 65 8D 4C 2E 29 0E 1E 26 4F BB F2 58 67 50 89 F1 45 BF 03 F6 5E D9
36 CB 8A D2 DB 77 3F D0 C0 B7 2B 85 2B 46 5C 6B 3B 64 BD 25 8A F1 33 18 AC 38 B1 69 F5 1E 07 23
53 DE A5 1B E1 80 44 79 5C EB 82 52 A2 51 94 90 D4 D1 E6 70 96 2C FE 29 50 C9 61 64 CE 27 F6 F1
00 71 CA DD F7 B2 12 15 D8 B5 0E 4E 11 90 E8 49 C2 62 1A A2 F2 62 22 A5 D3 F7 9A C0 7D 26 ED 51
AE 5B 80 AD 7A FA BC 28 E3 52 A7 E1 6D CD B9 8E 69 4D 5A 1A 7E 48 11 80 AB 4D 16 B7 A5 54 85
DA DF 78 94 B4 DD 1E 56 81 01 52 2A 7D 9A BF E8 B1 A6 A3 64 D5 E2 07 60 69 2E EB DC 8D 6A 4F
D8 21 4C 98 E5 B3 59 C4 B8 A5 00 B7 2E DE BB 3F BE E6 AF C1 B2 0A 2A 96 FA AA 92 30 9A 0A 72 1C
C2 B2 6A 54 BA 6A EA AD 14 34 5B CE 56 A3 DC 79 DD 4A 16 40 98 A8 FF EF 38 34 2D 62 FE 99 90
D0 F4 34 59 1F B8 A9 FE 94 64 84 DA 4A 70 EA 3B 14 FC 3A 08 57 F6 6D 05 12 CC F2 46 43 0D 9F 50
A7 BE 64 3E B2 5E CC 2C A9 F1 4A 22 54 2A 1D 6C 53 2F 54 6C F4 C1 27 99 6A 63 F4 7E 58 90 05 85
00 14 53 78 FC 3B 42 E8 0E 53 6A 25 33 B4 ED 8C 1D 7F A8 54 C8 AE 59 40 26 01 4E 7C 12 5F C1 DE
29 63 1F 5E B0 3F 3F B6 AC BB D0 88 B9 CA DA 47 7D 26 47 CF 80 36 03 78 A0 3D 2B 4D B0 F0 D2 77
18 87 D5 08 C8 88 E0 4B 24 EE AF A0 A0 B3 92 21 88 91 B2 20 BD 65 03 89 8E 80 B2 AB 13 AE D7 9C
28 AD 7B A1 92 6D 8F 52 22 72 E3 AE 5B A9 C1 26 5C 35 7C 0C E7 41 A6 5B 75 E2 14 04 64 C4 00 64
D2 91 76 E2 9A 67 6D 45 B6 7E E6 34 0A 6D D2 BA FB 9C 0C 80 04 1C 7C DC 3C 9C FC 71 CE 7E AE
75 A0 A0 A0 80 A0 A0 A0 D0 F5 BE D4 A7 92 B9 2C D2 DA BD BF 02 52 03 AC 3D B3 7A 09 1E 94 74
DA 4C 1E 21 FA 9E 9C EE 6E A0 81 76 FB E8 EF 45 CC 8C 9C 63 C4 95 6E 08 C8 0E 19 38 D2 97 A9 2B
0D BE C6 6B B1 51 32 18 90 B8 7F F2 33 38 C5 40 C2 EB A5 57 9A AE 13 06 D0 BB D0 08 78 78 C8 DF
4E 58 98 77 38 FB 04 9E FC AE 8E AE 1E BD CC 98 A8 4D B2 93 BC D5 FD 44 F1 B0 0D 1E C4 D0 17 D7
D0 09 D1 FC A4 FD 15 E8 9F 9F C8 C1 A9 26 F9 FE D0 BA 1D 3C 64 BD 25 48 3F 2E 26 4F EB 87 A8 5B
8D 10 60 24 58 90 62 3C AC 42 34 0D 0A

Upload: 26 26 3B 32 38 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 32 2C 00 00 00
00 04 00 01 31 44 0D 0A

Note: The address offset of the first sent data packet in the FOTA file is 00 00 00 00, the length of the sent FOTA data packet is 04 00, and the red data is the FOTA data packet.

Step 4: Repeat Step 3 until the entire FOTA file is downloaded:



File Name	iStartek GPS Tracker Protocol	Ver.	2.3
Update	2025-10-18	Page	78

Send: 24 24 3C 31 30 35 31 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 32 2C 00 00
04 00 04 00 F4 DC ED 96 3A 55 18 1A 25 9E 82 3D E8 20 40 98 1D 85 5E ED 46 CD 3D C6 EA A5 A0 18
94 63 61 63 46 36 74 B1 5C 3F 3F 28 21 C9 B5 CE 37 33 66 E1 B1 6A 33 7F 9E 2D CC 54 2D E5 20 D8
51 69 1C BB 0A E2 6B 67 92 95 3E B0 5F 8D 53 6C 45 AA EE 5A E6 6A 8C 9C 6E 48 4F 3E DE 0C BA C7
B0 00 62 14 A4 24 37 C6 22 02 59 14 78 15 72 6A DA 77 AA 83 FA DD FC A4 FD 05 18 7F 4E E6 0C 95
61 39 C8 F5 E6 A6 8B E0 07 3C 57 A4 04 D7 86 0E 77 13 6E E6 0F BB 1E 69 68 50 89 B1 4C 5B 3D 9C
6C 4E D0 E5 39 6A A4 BE 32 4E 94 ED B4 79 58 50 F4 50 00 48 BC 58 59 EB 1C 9D 71 44 0F D9 C1 DD
AC 4B BA 02 F6 82 6A BD D0 FC 41 36 DA E5 D9 32 AA 9D FC 64 B8 37 3A AD 3C CB 2A 7D BC C4 1E
0B 37 78 28 A7 FE B9 38 00 B9 61 1C 2B B2 B5 2C BB A2 25 3C CB 72 E7 FC AC 55 D6 7E C6 CE B3 E0
C5 0E 24 7A FA 59 A4 F8 3C EC E1 E3 79 C2 45 61 58 CA 38 3C 38 37 16 BC ED 2A F6 2F D8 C8 D6 38
49 AE 57 E1 22 72 53 55 06 D6 26 77 18 50 FA 9E 7D 4A 11 59 89 10 70 D0 77 6F F9 35 77 37 9E A2
AF 23 0C 4D 75 60 60 C0 01 12 EA 62 F4 88 4E D2 04 F5 B7 BA CA 5A CC 4A 42 26 63 2D FF 3A 8C 22
68 38 4F A6 5C 76 BA B0 8A 36 24 B8 33 CC 1B C6 3E D6 80 FC A3 18 BE 0B B9 8C EC 4A 30 44 31 AB
AD 4E 8E CE C9 B0 3A 04 F8 14 36 60 46 A3 AB FD 29 44 24 07 5E BA D7 C1 B3 EC 5C 71 66 46 81 28
E2 6C 90 AC CE 98 DE F9 3F B1 07 1F BE 0A 7C 3D 3F 82 D2 82 1A 52 A4 B8 9F 96 78 E5 9A A2 65 CD
5E 55 37 4E FE 29 FB C6 93 84 56 87 96 6A 8A 4D 4C E1 10 66 B0 E6 A6 D0 6E C6 77 27 96 09 78 98
F5 2A E4 4A F7 6F 6D 65 D6 D6 E3 78 62 3C AC F7 52 93 0C 0C 94 A4 5E F8 B0 40 48 E9 C2 A2 1A DA
E2 C2 3A 25 70 F1 F9 B7 EC C4 4A 12 99 A9 02 E2 DA 9A 22 02 FA 1E 8D 34 E5 46 4E 03 28 80 50 4F
B4 4C 02 8A E1 F1 EA B6 71 CC 14 E8 F7 8A 5D 0C 9B 60 85 44 F3 33 FC 16 E3 06 41 11 0A D3 DF 32
55 14 AC 46 B3 D6 F1 30 48 41 39 74 C3 1A E2 4A 0D 9C 77 DA 4A 7B B5 04 E2 2C BF 28 C8 30 E7 DE
BB 7F 0E 23 99 53 45 66 F7 DD 23 D1 8F E1 AE 8F 55 6D E0 61 84 B2 EC 7D 2B F1 EA 27 96 B6 1B 98
58 FD 4D D3 DD FC 74 3C 6C 3C 1C 0C 9B B2 4A 92 95 D4 4F 72 E2 A7 3A 7F 59 29 0D 42 36 D9 0E
D7 06 7F 64 9A BA BD 64 81 30 7E 36 76 33 D7 31 A0 55 1F D4 3C FC D3 02 42 14 E5 A4 B4 D3 25 64
74 9C 0B 1A 1A A1 AD 8E A9 31 2E 2C 6E 42 FA 43 80 E4 6C 05 12 06 5E 47 7C 80 68 61 F7 62 98 0F
42 F0 89 0B 2E 5E 2E B8 D1 A7 58 63 60 60 60 28 80 37 42 8D E0 C0 D0 B5 7A 29 29 3E 8A A2 E3 6C
A4 92 F7 5B 57 1E 11 7E 23 E4 61 CB 5D 32 95 57 69 5A 3A C7 24 E4 94 D9 B6 E3 5B C8 BE CE 82 92
82 7F F0 AC C8 7A F3 6A 1B E7 22 2B BE 02 EF 59 0C CE 75 EA 56 EE C7 58 78 86 0B 6F 12 18 86 AA
3E 82 93 49 FE C2 A4 94 04 84 74 F9 D6 03 74 C6 E9 D8 5E 3F 60 90 D9 7D DA A6 DC ED 9E A8 9E 3E
8A F1 73 48 67 7B FD 28 D1 F0 92 C2 C2 3F 2A BE 7C 02 D3 84 E4 A4 0D A9 88 4B B4 A0 FD 92 16 8C
74 7F 8B 48 84 D8 3D 6A AE 32 70 E3 C8 DB 77 5B 37 E0 1B A4 DD 2E 8E 93 D2 D4 64 49 3E B6 E6 77
D4 42 C7 9D 46 58 EA A6 40 10 90 F0 10 D8 A9 5E C3 DE 18 09 77 9F B2 B5 6C 49 AF 5A F4 87 BA 6B
64 6E 51 10 F3 30 59 D2 89 72 FD 2F BF 54 7F 51 E2 67 FD CE A3 C6 AC 59 BF 60 FB 40 AD 06 66 66
36 78 57 44 59 59 3A CC A3 2E C8 D8 E8 65 82 F7 87 12 D7 C4 0D 06 FB 88 25 9F A2 07 F7 6A 9C DC
2C 41 20 90 0F B4 42 34 0D 0A

Upload: 26 26 3C 32 38 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 32 2C 00 00 04
00 04 00 01 32 32 0D 0A

.....
.....

Send: 24 24 53 31 30 35 31 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 32 2C 00 01
74 00 04 00 BD E0 23 C8 41 F4 07 BC 75 80 FF 03 A2 6F EE 61 36 6F 16 12 A8 41 F4 53 B7 DE CB 22
3D 72 8B 93 E4 CB 20 DD D9 E6 54 DB 5C F4 EA AA E5 03 6F 86 98 37 90 80 56 15 D8 ED C7 68 C5 B0
29 10 D8 41 00 B1 AC 14 16 6A 5D 85 19 3C DD E4 D7 4E F7 DD 02 77 8E 3B 72 8B 90 78 ED 28 E1 88



File Name	iStartek GPS Tracker Protocol	Ver.	2.3
Update	2025-10-18	Page	80

Upload: 26 26 55 32 32 2C 38 36 35 32 33 35 30 35 30 30 31 37 31 34 30 2C 39 39 34 2C 01 32 46
0D 0A

Upload after successful firmware update: 26 26 3D 32 32 2C 38 36 35 32 33 35 30 35 30 30 31 37 31
34 30 2C 39 39 34 2C 02 31 38 0D 0A